

COMP90007 Internet Technologies
Project 1 – Network Analysis
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Section 2

Ans 2.1

Looking at Fig 2.1.1 and the result achieved in section 2 of the appendix Fig2.1.2

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	172.16.145.22	104.245.79.230	UDP	54	34161 → 33435 Len=12
2	0.002597	172.16.145.1	172.16.145.22	ICMP	82	Time-to-live exceeded (Time to live exceeded in transit)
3	0.002792	172.16.145.22	104.245.79.230	UDP	54	34161 → 33436 Len=12
4	0.028872	103.131.13.193	172.16.145.22	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
5	0.029179	172.16.145.22	104.245.79.230	UDP	54	34161 → 33437 Len=12
6	0.031978	103.251.4.88	172.16.145.22	ICMP	82	Time-to-live exceeded (Time to live exceeded in transit)
7	0.032173	172.16.145.22	104.245.79.230	UDP	54	34161 → 33438 Len=12
8	0.035034	103.251.4.31	172.16.145.22	ICMP	82	Time-to-live exceeded (Time to live exceeded in transit)
9	0.035298	172.16.145.22	104.245.79.230	UDP	54	34161 → 33439 Len=12
10	0.043778	202.139.7.25	172.16.145.22	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
11	0.044013	172.16.145.22	104.245.79.230	UDP	54	34161 → 33440 Len=12
12	2.049143	172.16.145.22	104.245.79.230	UDP	54	34161 → 33441 Len=12
13	4.051440	172.16.145.22	104.245.79.230	UDP	54	34161 → 33442 Len=12
14	6.056536	172.16.145.22	104.245.79.230	UDP	54	34161 → 33443 Len=12
15	6.075073	61.88.33.48	172.16.145.22	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
16	6.075390	172.16.145.22	104.245.79.230	UDP	54	34161 → 33444 Len=12
17	6.210602	203.208.147.113	172.16.145.22	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
18	6.210937	172.16.145.22	104.245.79.230	UDP	54	34161 → 33445 Len=12
19	6.352555	203.208.154.14	172.16.145.22	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
20	6.352882	172.16.145.22	104.245.79.230	UDP	54	34161 → 33446 Len=12
21	6.486519	184.104.194.90	172.16.145.22	ICMP	82	Time-to-live exceeded (Time to live exceeded in transit)
22	6.486850	172.16.145.22	104.245.79.230	UDP	54	34161 → 33447 Len=12
23	6.623334	203.208.166.61	172.16.145.22	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
24	6.623675	172.16.145.22	104.245.79.230	UDP	54	34161 → 33448 Len=12
25	8.626519	172.16.145.22	104.245.79.230	UDP	54	34161 → 33449 Len=12
26	10.631612	172.16.145.22	104.245.79.230	UDP	54	34161 → 33450 Len=12
27	11.119289	184.105.223.225	172.16.145.22	ICMP	82	Time-to-live exceeded (Time to live exceeded in transit)
28	11.119674	172.16.145.22	104.245.79.230	UDP	54	34161 → 33451 Len=12
29	11.403351	184.104.197.243	172.16.145.22	ICMP	82	Time-to-live exceeded (Time to live exceeded in transit)
30	11.403680	172.16.145.22	104.245.79.230	UDP	54	34161 → 33452 Len=12
31	11.733418	184.105.223.225	172.16.145.22	ICMP	82	Time-to-live exceeded (Time to live exceeded in transit)
32	11.733772	172.16.145.22	104.245.79.230	UDP	54	34161 → 33453 Len=12
33	12.055941	104.245.79.3	172.16.145.22	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
34	12.056224	172.16.145.22	104.245.79.230	UDP	54	34161 → 33454 Len=12
35	12.367806	104.245.79.110	172.16.145.22	ICMP	82	Time-to-live exceeded (Time to live exceeded in transit)
36	12.368132	172.16.145.22	104.245.79.230	UDP	54	34161 → 33455 Len=12
37	12.757409	104.245.79.3	172.16.145.22	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
38	12.757757	172.16.145.22	104.245.79.230	UDP	54	34161 → 33456 Len=12
39	13.068864	104.245.79.110	172.16.145.22	ICMP	82	Time-to-live exceeded (Time to live exceeded in transit)
40	13.069186	172.16.145.22	104.245.79.230	UDP	54	34161 → 33457 Len=12
41	13.474259	104.245.79.230	172.16.145.22	ICMP	82	Destination unreachable (Communication administratively filtered)

<Fig 2.1.1 Wireshark for one packet from my apartment to *speedtest.awbb.net*>

Traceroute probes the network path hop-by-hop by sending serialized UDP or ICMP messages with incrementing TTLs to the destination host. It uses the ICMP timeout messages returned by the router when the TTL expires to record the IP address and delay of each hop.

Therefore, we could use Wireshark to return ICMP protocol to obtain every hop's IP address. We also can get IP addresses directly from terminal.

My Wireshark capture filter is **icmp or (udp and portrange 33434-33534)**. We could divided it into two parts. The first part, **ICMP**, captures all ICMP protocol packets—this corresponds to the ICMP Time Exceeded message returned by the router during traceroute and the ICMP Destination Unreachable message returned by the final destination. Part 2, **UDP and portrange 33434-33534**. This port range covers 100 ports after the default starting port 33434 of traceroute, which is sufficient to capture all UDP probe packets in a normal traceroute session. They are sent to the target host in UDP protocol, with the destination port increasing hop by hop starting from 33434.

Will this filter capture non-traceroute traffic?

Yes. The ICMP part is quite broad—it captures not only traceroute-related ICMP Time Exceeded and Destination Unreachable errors, but also any other ICMP traffic. For example, if the user were pinging google.com while capturing packets, those ping ICMP packets will also be captured by this filter.

Similarly, for the UDP and port range 33434-33534, if other applications happen to use UDP communication within this port range.

From the Fig2.1, there are some information about the first 5 hops and last hop.

The 1 st hop	→	172.16.145.1 TTL (Time-to-live) = 1
The 2 nd hop	→	103.131.13.193 TTL = 2
The 3 rd hop	→	103.251.4.80 TTL = 3
The 4 th hop	→	103.251.4.31 TTL = 4
The 5 th hop	→	202.139.7.25 TTL = 5
The last hop	→	104.245.79.230 TTL = 23

We will analyse the 6 hops line by line.

The first hop is 172.16.145.1. It shows that the packet was sent to the private network gateway which is the student apartment IP address. So, we can't find any information by <https://db-ip.com/>. Because it's one private gateway.

The second hop is 103.131.13.193. According to section 2 of the appendix Fig2.1.3, We could search from <https://db-ip.com/> and find it belongs to Melbourne, [1] which maybe Metropolitan Area Network. This time's RTT is high up to 26.332ms, which means that the packet was transmitted to the local central network gateway.

The third hop is 103.251.4.80. According to section 2 of the appendix Fig2.1.4, We could search from <https://db-ip.com/> and find it belongs to Sydney.[1] Meanwhile, the RTT is quite low which means that we still send messages to the Australia network gateway.

The fourth hop is 103.251.4.31. According to section 2 of the appendix Fig2.1.5, We could search from <https://db-ip.com/> and find it belongs to Sydney. [1] Similarly, the RTT indicates that we still send messages to the Australia network gateway. Meanwhile, the IP address there is a difference only in the last byte, which means that both belong to one Internet Service Provider (ISP).

The fifth hop is 202.139.7.25. According to section 2 of the appendix Fig2.1.6, We could search from <https://db-ip.com/> and find it belongs to Sydney. [1] However, in this case, the RTT suddenly increased significantly from 3ms to 8ms, which may mean the network is preparing for the international transmission. The packet arrived at the peering point. Then, the following traceroute, there are 3 following hops exceed time which is also proving that the result.

In summary, until the fifth hop, all RTT is under 9ms, which is low compared with the future RTT. We could know until now that we still send the ICMP messages in Australia.

The last hop is 104.245.79.230. We can see it in Wireshark. There is one info which shows Destination unreachable. This means our request has reached the destination. No matter whether it's reachable or not, it's the IP address of [speedtest.awbb.net](https://www.speedtest.net/). According to RTT number, we could easily know we sent the ICMP message to one international network gateway.

Ans 2.2

It continues section 2 of the appendix Fig 2.2.4~Fig 2.2.13[1]

I choose the iperf.he.net as the 10th server which is in California, USA.

We could get the latitude and longitude of the ten servers. And there are some interesting finds. Firstly, the third server (speedtest.wobcom.de) in Austria is actually in Germany [1] and in the same city which has 2nd server (spd-desrv.hostkey.com). Then, the 7th server is listed in Porto, Portugal, but it locates in Lisbon in fact. [1] The last some server can't use UDP traceroute mode, therefore ICMP traceroute mode is an available alternative.

And then, I used one algorithm to calculate the distance between each server and Melbourne CBD. The great-circle distances between Melbourne and each destination were computed using the spherical law of cosines formulation of the Haversine distance (Sinnott, 1984), with a mean Earth radius of 6371 km.[2]

According to Fig 2.2 in section 2 of appendix, destinations' latitude and longitude come from the <https://db-ip.com/>. [1] Depending on a python calculator – Fig 2.2.3 in appendix for section 2, we can get the result of the distance.

The distance between two points on the Earth's surface (assumed spherical) is computed with the Haversine spherical law of cosines formula:[2]

$$d = R * \arccos(\sin(\varphi_1) * \sin(\varphi_2) + \cos(\varphi_1) * \cos(\varphi_2) * \cos(\lambda_2 - \lambda_1))$$

φ_1, λ_1 | Latitude and longitude → origin (Melbourne)

φ_2, λ_2 | Latitude and longitude → destination

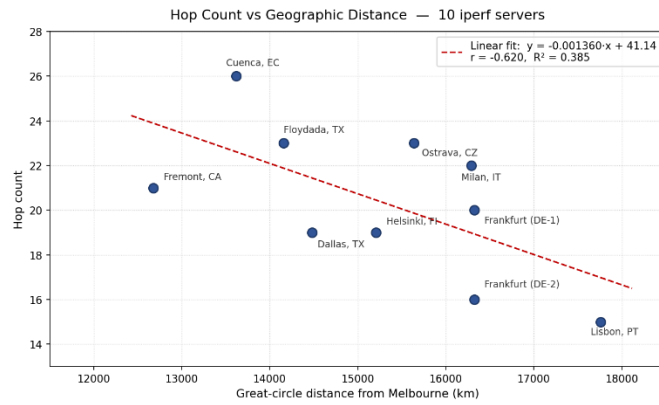
There is one interesting find that except the 10th server I choose by myself, the farthest server is 7th server(porto.speedtest.net.zon.pt) in Portugal and the closest server is 8th server(speedtest.masnet.ec) in Ecuador. The two server we will use in the section 4.

And make a summary -- Fig 2.2.1

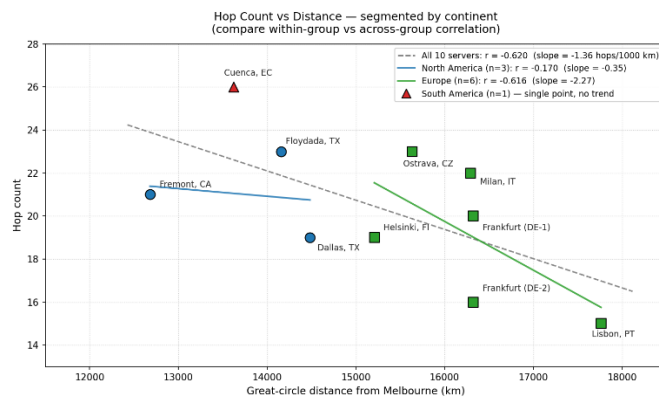
#	Hostname	Destination IP	Listed Region	Traceroute Mode	Hop Count	Detected City	Country	Latitude	Longitude	Distance (km)	Notes
Origin	MY LOCATION (Melbourne CBD)	—	Australia	—	0	Melbourne	Australia	-37.8136	144.9631	0	Reference point for Haversine distance
1	speedtest.lawbb.net	104.245.79.230	USA	UDP	23	Floydada	United States	33.9845	-101.338	14155	
2	spd-desrv.hostkey.com	185.84.224.68	Germany	ICMP	20	Frankfurt am Main	Germany	50.1109	8.68213	16321	
3	speedtest.wobcom.de	62.176.224.217	Austria (listed) / Germany?	ICMP	16	Frankfurt am Main	Germany	50.1109	8.68213	16321	Listed as Austria; IP geolocates to Frankfurt, DE
4	rychlost.poda.cz	85.135.13.2	Czech Republic	UDP	23	Ostrava	Czechia	49.8314	18.2776	15636	
5	spd-fisrv.hostkey.com	212.237.217.41	Finland	ICMP	19	Helsinki	Finland	60.1699	24.9384	15207	
6	speed.itgate.net	213.212.129.36	Italy	ICMP	22	Milan	Italy	45.4642	9.18998	16292	
7	porto.speedtest.net.zon.pt	95.95.252.95	Portugal	ICMP	15	Lisbon	Portugal	38.7599	-9.15765	17761	Hostname "porto" but IP geolocates to Lisbon
8	speedtest.masnet.ec	170.83.216.19	Ecuador	UDP	26	Santa Ana (Cuenca)	Ecuador	-2.94017	-78.92	13621	
9	daf.speedtest.clouvider.net	2.56.188.136	USA (Dallas)	UDP	19	Dallas	United States	32.7892	-96.8217	14482	
10	iperf.he.net	216.218.207.42	USA (Fremont, CA) — Hurricane Electric	UDP	21	Fremont	United States	37.49	-121.931	12679	Hurricane Electric (replaced Auckland; iperf service still live)

<Fig 2.2.1 the summary for 10 iperf servers >

Therefore, we could use the table to make A scatter chart of the hop count versus the approximate geographical distance from Melbourne.



<Fig 2.2.2(a) A scatter chart of the hop count versus the approximate geographical distance from Melbourne >



<Fig 2.2.2(b) A scatter chart by continent>

From Fig 2.2.2(a), we can find $R^2 = 0.385$ and $r = -0.620$, with the linear fit $y = -0.01360x + 41.14$. For the hosts measured in this experiment, hop count tends to decrease with the approximate geographical distance increasing. This indicates a moderate negative correlation in this dataset.

There are two main reasons. First, long-distance routes often utilise large backbone networks with fewer intermediate routing domains, whereas shorter routes may still traverse multiple peering points or smaller ISPs. For example, the Ecuador host is geographically much closer than several European hosts, but its hop count is really high, which suggests that the path to South America passes through more intermediate networks.

Second, hop count reflects topological distance rather than purely geographical distance. Two geographically distant nodes connected through well-optimised backbone links may still have a relatively low hop count, while geographically closer nodes can show a higher hop count if the route crosses more Autonomous Systems or peering boundaries [3, p.396].

From Fig 2.2.2(b), after grouping the hosts by continent, the six European servers show a negative correlation ($r = -0.616$), while the three North American servers show almost no correlation ($r = -0.170$). This suggests that the observed relationship between hop count and distance may depend more on route structure, peering arrangements, and backbone usage than on geographical distance alone.

Section 3

Ans 3.1

Looking at Fig 3.1.1 and the result achieved in section 3 of the appendix Fig 3.1.2

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	172.16.145.22	104.245.79.230	ICMP	98	Echo (ping) request id=0xd253, seq=0/0, ttl=64 (reply in 2)
2	0.309789	104.245.79.230	172.16.145.22	ICMP	98	Echo (ping) reply id=0xd253, seq=0/0, ttl=48 (request in 1)
3	1.009953	172.16.145.22	104.245.79.230	ICMP	98	Echo (ping) request id=0xd253, seq=1/256, ttl=64 (reply in 4)
4	1.318832	104.245.79.230	172.16.145.22	ICMP	98	Echo (ping) reply id=0xd253, seq=1/256, ttl=48 (request in 3)
5	2.010283	172.16.145.22	104.245.79.230	ICMP	98	Echo (ping) request id=0xd253, seq=2/512, ttl=64 (reply in 6)
6	2.324469	104.245.79.230	172.16.145.22	ICMP	98	Echo (ping) reply id=0xd253, seq=2/512, ttl=48 (request in 5)
7	3.014535	172.16.145.22	104.245.79.230	ICMP	98	Echo (ping) request id=0xd253, seq=3/768, ttl=64 (reply in 8)
8	3.406793	104.245.79.230	172.16.145.22	ICMP	98	Echo (ping) reply id=0xd253, seq=3/768, ttl=48 (request in 7)

< Fig 3.1.1 Wireshark for ping speedtest.awbb.net>

Looking at Fig 3.1.1, my machine sends 4 ICMP echo request (type: Echo (ping) request (8), code: 0) to the destination, we could see them in the picture – No.1, No.3, No.5, No.7. The destination will send ping reply (type: Echo (ping) reply (0), code: 0). The original TTL is 64 and the reply TTL is 48, which means that the packets traverses $64 - 48 = 16$ hops. And we can see that all ICMP id is 0xd253 which indicates the 4 request and 4 reply belong to one time ping command.

Turn to Fig 3.1.2, the data bytes from speedtest.awbb.net is 56 bytes. However, each packet was transmitted 64 bytes, because ICMP header is 8 bytes which are included in the 64 bytes transition. And we could see the icmp_seq = 0/1/2/3[4] which represents 4 ICMP packets were sent from speedtest.awbb.net.

I used the icmp and host speedtest.awbb.net as the capture filter. By capturing only the ICMP traffic between our machine and this specific target, and filtering out all other irrelevant packets, the packet capture results are much cleaner. However, it can capture other requests (non-ping related), such as traceroute speedtest.awbb.net.

Ans 3.2

Looking at Fig 3.2.1 ~ 3.2.10

Make a table to summarize all information.

Host #	Hostname (requested)	Resolved Host	IP	Run #	Packets TX	Packets RX	Packet Loss %	min (ms)	avg (ms)	max (ms)	stddev (ms)
1	speedtest.awbb.net	speedy.awbb.net	104.245.79.230	1	4	4	0.0%	323.385	406.673	489.509	73.655
1	speedtest.awbb.net	speedy.awbb.net	104.245.79.230	2	4	4	0.0%	310.689	342.574	373.521	23.167
1	speedtest.awbb.net	speedy.awbb.net	104.245.79.230	3	4	4	0.0%	323.581	357.839	401.844	28.970
2	spd-desrv.hostkey.com	spd-desrv.hostkey.com	185.84.224.68	1	4	4	0.0%	280.900	331.869	389.726	41.383
2	spd-desrv.hostkey.com	spd-desrv.hostkey.com	185.84.224.68	2	4	4	0.0%	284.299	292.768	310.441	10.637
2	spd-desrv.hostkey.com	spd-desrv.hostkey.com	185.84.224.68	3	4	4	0.0%	313.697	342.068	370.486	21.152
3	speedtest.wobcom.de	speedtest.wobcom.de	62.176.224.217	1	4	4	0.0%	303.360	329.810	363.234	25.298
3	speedtest.wobcom.de	speedtest.wobcom.de	62.176.224.217	2	4	4	0.0%	302.182	316.028	338.643	14.850
3	speedtest.wobcom.de	speedtest.wobcom.de	62.176.224.217	3	4	4	0.0%	302.931	326.152	379.141	30.899
4	rychlost.poda.cz	rychlost.poda.cz	85.135.13.2	1	4	4	0.0%	325.757	354.667	383.288	21.489
4	rychlost.poda.cz	rychlost.poda.cz	85.135.13.2	2	4	4	0.0%	336.553	368.015	398.892	23.726
4	rychlost.poda.cz	rychlost.poda.cz	85.135.13.2	3	4	4	0.0%	309.479	361.332	397.973	32.958
5	spd-fisrv.hostkey.com	spd-fisrv.hostkey.com	212.237.217.41	1	4	4	0.0%	294.590	320.340	360.514	25.228
5	spd-fisrv.hostkey.com	spd-fisrv.hostkey.com	212.237.217.41	2	4	4	0.0%	312.455	349.619	395.802	30.498
5	spd-fisrv.hostkey.com	spd-fisrv.hostkey.com	212.237.217.41	3	4	4	0.0%	330.131	359.502	389.968	22.133
6	speed.itgate.net	speed.itgate.net	213.212.129.36	1	4	4	0.0%	308.254	346.621	394.541	38.355
6	speed.itgate.net	speed.itgate.net	213.212.129.36	2	4	4	0.0%	307.373	346.364	380.658	26.853
6	speed.itgate.net	speed.itgate.net	213.212.129.36	3	4	4	0.0%	307.918	346.577	384.572	30.337
7	porto.speedtest.net.zon.pt	porto.speedtest.net.zon.pt	95.95.252.95	1	4	4	0.0%	267.539	299.773	342.463	27.713
7	porto.speedtest.net.zon.pt	porto.speedtest.net.zon.pt	95.95.252.95	2	4	4	0.0%	268.110	313.233	366.067	43.912
7	porto.speedtest.net.zon.pt	porto.speedtest.net.zon.pt	95.95.252.95	3	4	4	0.0%	283.659	342.310	450.399	64.395
8	speedtest.masnet.ec	speedtest.masnet.ec	170.83.216.19	1	4	4	0.0%	325.073	358.571	430.035	42.076
8	speedtest.masnet.ec	speedtest.masnet.ec	170.83.216.19	2	4	4	0.0%	325.178	374.309	424.896	41.764
8	speedtest.masnet.ec	speedtest.masnet.ec	170.83.216.19	3	4	4	0.0%	325.110	328.279	337.505	5.327
9	dal.speedtest.clouvider.net	dal.speedtest.clouvider.net	2.56.188.136	1	4	4	0.0%	190.588	192.087	195.053	1.743
9	dal.speedtest.clouvider.net	dal.speedtest.clouvider.net	2.56.188.136	2	4	4	0.0%	192.074	195.067	196.429	1.759
9	dal.speedtest.clouvider.net	dal.speedtest.clouvider.net	2.56.188.136	3	4	4	0.0%	190.827	191.965	192.534	0.688
10	iperf.he.net	1500.mtu.he.net	216.218.207.42	1	4	4	0.0%	296.809	385.403	475.908	79.867
10	iperf.he.net	1500.mtu.he.net	216.218.207.42	2	4	4	0.0%	287.294	303.400	341.647	22.290
10	iperf.he.net	1500.mtu.he.net	216.218.207.42	3	4	4	0.0%	282.754	309.932	353.514	28.101

< Fig 3.2.11 The summary of 30 times ping to 10 servers>

And I used Function STDEV.S which is built into the Excel file. STDEV.S implements the sample standard deviation formula $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 / (N - 1)]}$, where $N = 12$.

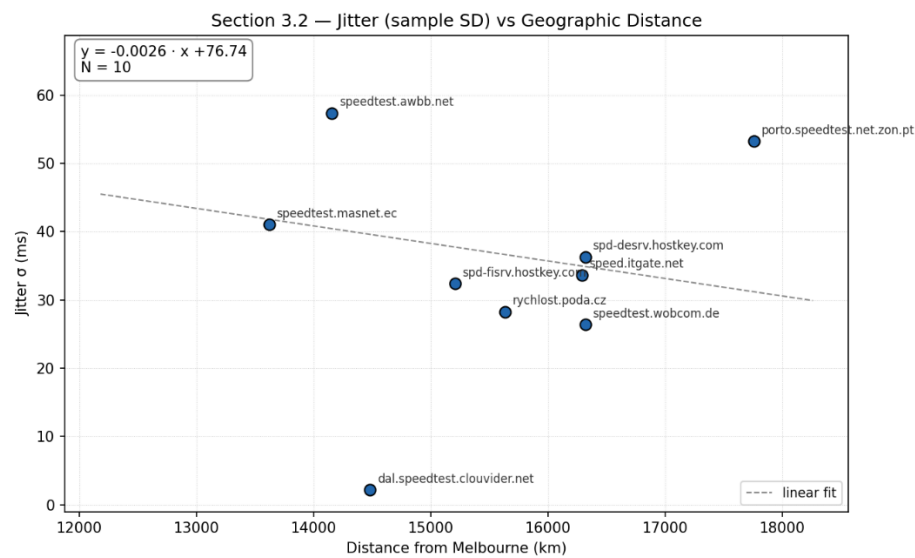
Section 3.2 — Per-Host Summary (N = 12 packets per host = 3 runs × 4 echoes)

Mean RTT $\bar{x} = \sum x_i / N$ Jitter $\sigma = \sqrt{[\sum (x_i - \bar{x})^2] / (N - 1)}$ Distance: from Section2.2_Analysis.xlsx → 'Distance vs Hops' sheet, column K

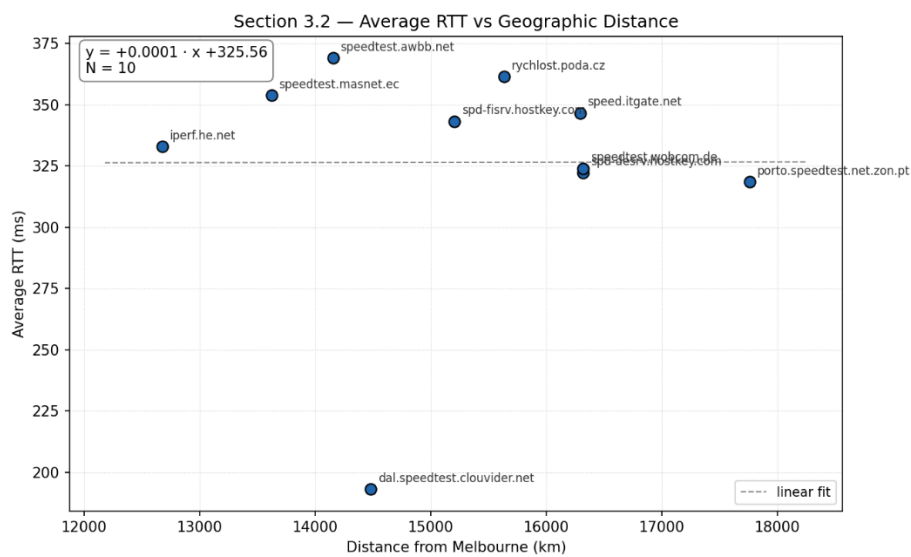
Host #	Hostname	IP	Distance (km)	Avg RTT (ms)	Jitter σ (ms)	N
1	speedtest.awbb.net	104.245.79.230	14,155	369.028	57.345	12
2	spd-desrv.hostkey.com	185.84.224.68	16,321	322.235	36.320	12
3	speedtest.wobcom.de	62.176.224.217	16,321	323.996	26.404	12
4	rychlost.poda.cz	85.135.13.2	15,636	361.338	28.285	12
5	spd-fisrv.hostkey.com	212.237.217.41	15,207	343.154	32.395	12
6	speed.itgate.net	213.212.129.36	16,292	346.521	33.643	12
7	porto.speedtest.net.zon.pt	95.95.252.95	17,761	318.439	53.218	12
8	speedtest.masnet.ec	170.83.216.19	13,621	353.720	41.066	12
9	dal.speedtest.clouvider.net	2.56.188.136	14,482	193.039	2.155	12
10	iperf.he.net	216.218.207.42	12,679	332.911	65.559	12

< Fig 3.2.12 Per-Host Summary (distance & jitter)>

According to the above table, we could easily Plot the scatter charts for distance VS jitter and distance VS avg RTT.



< Fig 3.2.13 Scatter chart (distance VS jitter)>



< Fig 3.2.14 Scatter chart (distance VS Avg RTT)>

Ans 3.3

Fig. 3.2.13 suggests a weak negative relationship between physical distance and jitter. We could say that more distant hosts do not necessarily have higher jitter, and the downward trend is not strong. The presence of a very stable host such as `dal.speedtest.clouvider.net` also indicates that route-specific conditions can reduce the effect of distance.

According to Fig 3.2.14, the fitted slope is approximately zero and the trendline is nearly horizontal. This indicates almost no linear correlation between average RTT and geographical distance in these measurements. Therefore, average RTT here appears to be influenced more by routing path quality, queueing, peering, and endpoint or access-network conditions than by distance alone.

All measurements were conducted over a shared Wi-Fi network in a student apartment building, where the wireless access point serves many concurrent users across the entire building. The typical download speed observed on this network is approximately 80 Mbps. During evening peak hours, when residents were streaming video, gaming, and downloading files simultaneously, noticeable increases in latency and network instability were observed.

This shared wireless environment has two significant implications for the results. First, the Wi-Fi access point can introduce additional queueing delay under load, which is added to every ping measurement regardless of the destination. This local overhead raises the baseline RTT for all hosts and reduces the relative contribution of pure propagation delay, helping to explain why RTT shows virtually no correlation with distance. Second, fluctuations in local network load can contribute to jitter variation. For example, `speedtest.masnet.ec` recorded a jitter of 42.076ms in Run 1 and 41.764ms in Run 2, but only 5.327ms in Run 3. This pattern is consistent with changing local congestion, although server-side load and path conditions may also have contributed. Similarly, `iperf.he.net` recorded an RTT spike of 475.908ms in Run 1, which is also consistent with a short-lived congestion event on either the local access network or the wider path.

However, local network conditions alone do not account for all observed variation. `dal.speedtest.clouvider.net` had both the lowest RTT (193.039ms) and the lowest jitter (2.155ms) among all 10 hosts, with highly consistent results across all three runs. This suggests that when the end-to-end route is well optimised, stable performance is achievable even over a shared Wi-Fi connection. In contrast, `speedtest.awbb.net`, located at a similar distance (14,155 km vs 14,482 km), recorded an RTT of 369.028ms and a jitter of 57.345ms. This large difference for two hosts at comparable distances suggests that routing path quality and intermediate network conditions were more influential than geographical distance in determining delay and jitter in this experiment.

Section 4

Ans 4.1

Firstly, bandwidth-delay product (BDP) is defined as the product of a link's bandwidth and the round-trip delay time [5]. It represents the maximum amount of data that can be in transit on the network at any given moment. A higher BDP indicates that the network path has a larger capacity for in-flight data, which is important for TCP performance — the TCP send window must be at least as large as the BDP to fully utilise the available bandwidth [5]. In summary, the BDP determines the minimum size required for a TCP send window. If the send window is smaller than the BDP, the sender cannot continue sending data while waiting for the

ACK, resulting in link utilization falling below 100% and actual throughput being far lower than the available link bandwidth.

Then, we look at Fig 4.1(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) in section 3 of appendix and make a table to summarize the data in the under Fig 4.1.1 & Fig 4.1.2

Host #	Hostname	Resolved IP	Country/Region	Run #	Command	Interval (sec)	Transfer (MBytes)	Bandwidth (Mbps/sec)	Notes
1	speedtest.awbb.net	104.245.79.230	USA (TX)	1	iperf -c speedtest.awbb.net -p 80	0.00-10.01	85.00	71.30	
1	speedtest.awbb.net	104.245.79.230	USA (TX)	2	iperf -c speedtest.awbb.net -p 80	0.00-10.01	48.80	40.90	
1	speedtest.awbb.net	104.245.79.230	USA (TX)	3	iperf -c speedtest.awbb.net -p 80	0.00-10.01	86.00	72.10	
2	spd-desrv.hostkey.com	185.84.224.68	Germany	1	iperf -c spd-desrv.hostkey.com -p 80	0.00-10.01	105.00	88.30	
2	spd-desrv.hostkey.com	185.84.224.68	Germany	2	iperf -c spd-desrv.hostkey.com -p 80	0.00-10.01	101.00	84.50	
2	spd-desrv.hostkey.com	185.84.224.68	Germany	3	iperf -c spd-desrv.hostkey.com -p 80	0.00-10.02	98.50	82.50	
3	speedtest.wobcom.de	62.176.224.217	Germany	1	iperf -c speedtest.wobcom.de -p 80	0.00-10.00	86.60	72.60	
3	speedtest.wobcom.de	62.176.224.217	Germany	2	iperf -c speedtest.wobcom.de -p 80	0.00-10.01	93.10	78.10	
3	speedtest.wobcom.de	62.176.224.217	Germany	3	iperf -c speedtest.wobcom.de -p 80	0.00-10.01	90.90	76.20	
4	rychlost.poda.cz	85.135.13.2	Czech Republic	1	iperf -c rychlost.poda.cz -p 80	0.00-10.02	69.60	58.30	
4	rychlost.poda.cz	85.135.13.2	Czech Republic	2	iperf -c rychlost.poda.cz -p 80	0.00-10.00	64.40	54.00	
4	rychlost.poda.cz	85.135.13.2	Czech Republic	3	iperf -c rychlost.poda.cz -p 80	0.00-10.02	4.75	3.98	Anomalous low throughput — likely server-side congestion
5	spd-fisrv.hostkey.com	212.237.217.41	Finland	1	iperf -c spd-fisrv.hostkey.com -p 80	0.00-10.03	91.60	76.60	
5	spd-fisrv.hostkey.com	212.237.217.41	Finland	2	iperf -c spd-fisrv.hostkey.com -p 80	0.00-10.00	64.40	54.00	
5	spd-fisrv.hostkey.com	212.237.217.41	Finland	3	iperf -c spd-fisrv.hostkey.com -p 80	0.00-10.03	51.50	43.10	
6	speed.itgate.net	213.212.129.36	Italy	1	iperf -c speed.itgate.net -p 80	0.00-11.26	1.50	1.12	Anomalous: connection took 11.26s to transfer only 1.50
6	speed.itgate.net	213.212.129.36	Italy	2	iperf -c speed.itgate.net -p 80	0.00-10.05	17.60	14.70	
6	speed.itgate.net	213.212.129.36	Italy	3	iperf -c speed.itgate.net -p 80	0.00-10.01	100.00	84.00	
7	porto.speedtest.net.zon.pt	95.95.252.95	Portugal (Lisbon)	1	iperf -c porto.speedtest.net.zon.pt -p 80	0.00-10.04	16.60	13.90	
7	porto.speedtest.net.zon.pt	95.95.252.95	Portugal (Lisbon)	2	iperf -c porto.speedtest.net.zon.pt -p 80	0.00-10.14	16.50	13.70	
7	porto.speedtest.net.zon.pt	95.95.252.95	Portugal (Lisbon)	3	iperf -c porto.speedtest.net.zon.pt -p 80	0.00-10.05	16.40	13.70	
8	speedtest.masnet.ec	170.83.216.19	Ecuador	1	iperf -c speedtest.masnet.ec -p 80	0.00-10.34	14.00	11.40	
8	speedtest.masnet.ec	170.83.216.19	Ecuador	2	iperf -c speedtest.masnet.ec -p 80	0.00-10.34	23.00	18.70	
8	speedtest.masnet.ec	170.83.216.19	Ecuador	3	iperf -c speedtest.masnet.ec -p 80	0.00-10.34	15.50	12.60	
9	dal.speedtest.clouvider.net	2.56.188.136	USA (Dallas)	1	iperf -c dal.speedtest.clouvider.net -p 80	0.00-10.01	95.40	79.90	
9	dal.speedtest.clouvider.net	2.56.188.136	USA (Dallas)	2	iperf -c dal.speedtest.clouvider.net -p 80	0.00-10.00	123.00	103.00	
9	dal.speedtest.clouvider.net	2.56.188.136	USA (Dallas)	3	iperf -c dal.speedtest.clouvider.net -p 80	0.00-10.02	133.00	111.00	
10	iperf.he.net	216.218.207.42	USA (Fremont)	1	iperf3 -c iperf.he.net -R	0.00-10.01	43.00	36.10	Reverse mode (receiver); forward mode refused
10	iperf.he.net	216.218.207.42	USA (Fremont)	2	iperf3 -c iperf.he.net -R	0.00-10.00	33.40	28.00	Reverse mode (receiver)
10	iperf.he.net	216.218.207.42	USA (Fremont)	3	iperf3 -c iperf.he.net -R	0.00-10.01	18.40	15.40	Reverse mode (receiver)

<Fig 4.1.1 summary for 30times iperf command>

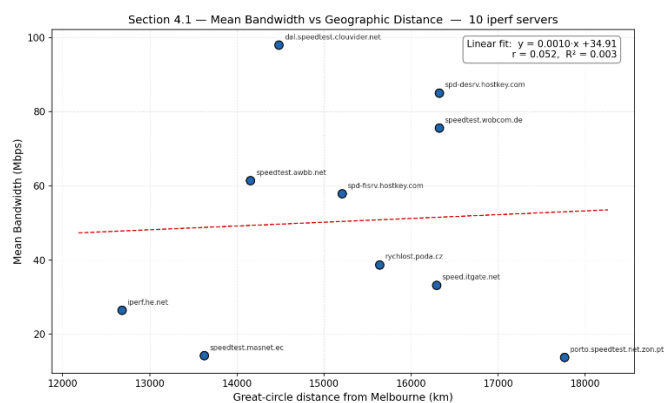
In Fig 4.1.1, there is something need to pay attention in Host 10, since direct access via forward mode was consistently denied, a receiver-side approach was used to read data, i.e., `iperf3 -c iperf.he.net -R`.

Host #	Hostname	Country/Region	Distance (km)	Run1 (Mbps)	Run2 (Mbps)	Run3 (Mbps)	Mean (Mbps)	StdDev (Mbps)	Min (Mbps)	Max (Mbps)	Notes
1	speedtest.awbb.net	USA (TX)	14,155	71.30	40.90	72.10	61.43	17.79	40.90	72.10	
2	spd-desrv.hostkey.com	Germany	16,321	88.30	84.50	82.50	85.10	2.95	82.50	88.30	
3	speedtest.wobcom.de	Austria (Germany)	16,321	72.60	78.10	76.20	75.63	2.79	72.60	78.10	
4	rychlost.poda.cz	Czech Republic	15,636	58.30	54.00	3.98	38.76	30.20	3.98	58.30	Run3 anomalous (3.98 Mbps) retained as-is
5	spd-fisrv.hostkey.com	Finland	15,297	76.60	54.00	43.10	57.90	17.09	43.10	76.60	
6	speed.itgate.net	Italy	16,292	1.12	14.70	84.00	33.27	44.45	1.12	84.00	Run1 anomalous (1.12 Mbps) retained as-is
7	porto.speedtest.net.zon.pt	Portugal (Lisbon)	17,761	13.90	13.70	13.70	13.77	0.12	13.70	13.90	[4.2 Furthest host]
8	speedtest.masnet.ec	Ecuador	13,621	11.40	18.70	12.60	14.23	3.91	11.40	18.70	[4.2 Closest host]
9	dal.speedtest.clouvider.net	USA (Dallas)	14,482	79.90	103.00	111.00	97.97	16.15	79.90	111.00	
10	iperf.he.net	USA (Fremont)	12,679	36.10	28.00	15.40	26.50	10.43	15.40	36.10	Reverse-mode values (server forward mode refused)

<Fig 4.1.2 summary for 30times iperf command>

From Fig 4.1.1 & Fig4.1.2, we could easily find the highest average bandwidth in the ten hosts is 97.97Mbps which belongs to dal.speedtest.clouvider.net. But it's not the closest host among all the hosts. And the lowest average bandwidth is 13.77Mbps, which belongs to porto.speedtest.net.zon.pt. This is the furthest host among all.

Therefore, we use Mean Bandwidth (Mbps) and distance to fit a linear trendline. The last, Host 4 and Host 6 each had one abnormally low value (3.98 and 1.12 Mbps respectively), which may have been caused by momentary server congestion. Unfortunately, it is reported that there is no display, but it is certain that it also exists and when using forward mode, iperf is unable to connect successfully and the number of returns is fixed.



<Fig 4.1.3 Scatter chart and fitted linear for mean Bandwidth VS Distance>

According to Fig 4.1.3, the regression line has a slightly positive slope, but the R^2 is only about 0.003. This indicates essentially no meaningful linear correlation between mean bandwidth and geographical distance in this dataset. Therefore, it would not be rigorous to conclude that bandwidth increases with distance; instead, the results suggest that server capacity, transient congestion, and path conditions are much more important than distance alone.

Ans 4.2

Looking at Fig 4.2.1 and Fig 4.2.2 in section 4 of the appendix

Host	Role	Distance (km)	Slot	Timestamp (AEST)	iperf Bandwidth (Mbps)	iperf Transfer (MBytes)	Interval (s)	ceroute H	Final-hop RTT (ms)
porto.speedtest.net.zon.pt	Furthest	17761	Morning	24 Apr 2026 09:22 AEST	13.8	16.5	0.00-10.03	15	269
porto.speedtest.net.zon.pt	Furthest	17761	Afternoon	24 Apr 2026 13:15 AEST	13.4	16	0.00-10.03	15	268.9
porto.speedtest.net.zon.pt	Furthest	17761	Evening	24 Apr 2026 18:02 AEST	13.8	16.5	0.00-10.01	15	268.3
porto.speedtest.net.zon.pt	Furthest	17761	Night	24 Apr 2026 23:02 AEST	13.6	16.3	0.00-10.03	15	301.8
speedtest.masnet.ec	Closest	13621	Morning	24 Apr 2026 09:25 AEST	20.6	25.4	0.00-10.34	25	327.9
speedtest.masnet.ec	Closest	13621	Afternoon	24 Apr 2026 13:18 AEST	22.2	27.4	0.00-10.34	25	324.6
speedtest.masnet.ec	Closest	13621	Evening	24 Apr 2026 18:09 AEST	24.4	30.1	0.00-10.34	25	326.1
speedtest.masnet.ec	Closest	13621	Night	24 Apr 2026 23:08 AEST	20.8	25.6	0.00-10.34	25	326.5

<Fig 4.2.3 Table for the Furthest and closest iperf servers in the whole day >

Porto (the farthest location, 17,761 km away) demonstrated remarkably stable throughput across all four time slots, with fluctuations ranging only from 13.4 to 13.8 Mbps (Upper and lower limits are 0.4 Mbps). The trace path consistently remained at 15 hops, and its RTT also stabilized at 268–270 milliseconds during morning, afternoon and evening. At night, although the RTT latency increased by about 30ms, the bandwidth remained relatively stable.

Ecuador (the closest location, 13,621 km away) exhibits greater bandwidth variability, with throughput fluctuating between 20.6 and 24.4 Mbps (Upper and lower limits are 3.8 Mbps), approximately nine times the upper and lower limits observed in Porto. There are two interesting findings. First, bandwidth peaked during the evening (AU), reaching 24.4 Mbps. Second, during the night, bandwidth dropped to 20.8 Mbps, while RTT surged from approximately 326 milliseconds to 413 milliseconds. The traceroute results suggest a possible factor: RTT at hop 24 (voip.masnet.ec) fluctuates significantly, whereas the morning traceroute results show a more stable RTT for the same hop. A possible reason is the time zone problem: 18:09 AEST corresponds roughly to 03:00 local time in Ecuador, which is likely an off-peak period with lower destination load. However, nighttime in Australia corresponds to peak network usage times in the Americas. Therefore, it's reasonable to speculate that the time zone difference causes the different RTT latency.

Slot	Portugal (Furthest)	Ecuador (Closest)
Morning	13.80	20.60
Afternoon	13.40	22.20
Evening	13.80	24.40
Night	13.60	20.80
Mean (Mbps)	13.65	22.00
StdDev (Mbps)	0.19	1.75
Min	13.40	20.60
Max	13.80	24.40
Range	0.40	3.80

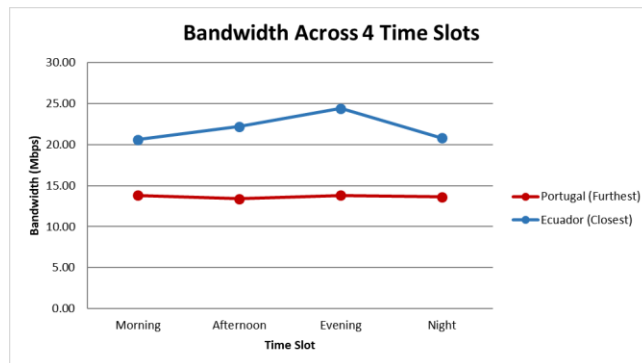
<Fig 4.2.4 Summary for the Furthest and closest iperf server in the whole day >

From Fig 4.2.4, the data shows that although the Avg bandwidth of the furthest server in Portugal, is only about half that of the nearest server in Ecuador, it remains relatively stable throughout all four time periods of the day. We used the STDEV algorithm, which is included in EXCEL file, get the Porto's SD (0.19 Mbps) and Ecuador's (1.75 Mbps). The former is one-tenth of the latter.

Overall, Porto appears to be constrained by a stable low-rate bottleneck near the destination, so time-of-day changes have little visible effect on throughput. Ecuador, by contrast, does not show the same fixed cap and

appears to be more sensitive to changing path conditions and destination-side usage patterns. The traceroute evidence suggests that one intermediate node may have contributed to the night-time degradation, but this should be treated as a plausible interpretation rather than definitive proof.

We could plot one chart to show the result more clearly.



<Fig 4.2.5 Chart for the Furthest and closest iperf servers in the whole day>

Ans 4.3

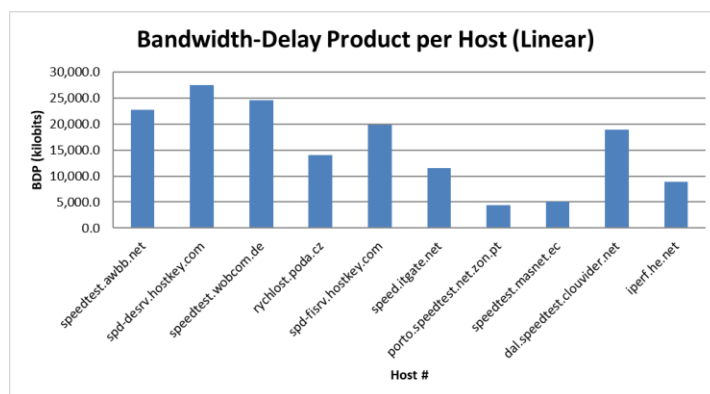
We used the data from section 3.2 and 4.1. Then we multiplied them with each other to get BDP.

BDP = mean RTT * mean Bandwidth

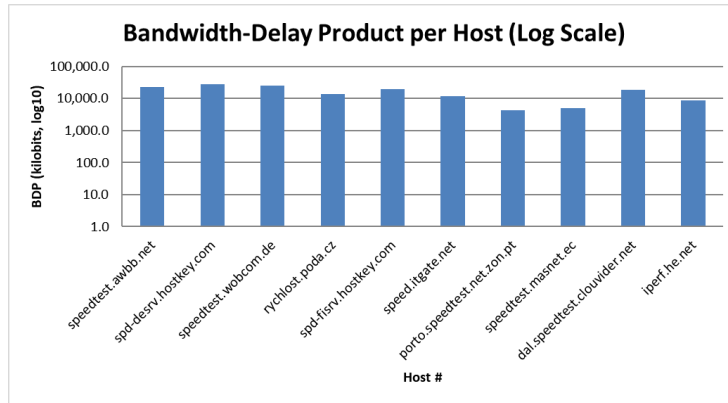
Host #	Hostname	Distance (km)	Mean RTT (ms) [from S3.2]	Mean Bandwidth (Mbps) [from S4.1]	BDP (kilobits)	BDP (KBytes)
1	speedtest.awbb.net	14,155	369.03	61.43	22,670.6	2,767.4
2	spd-desrv.hostkey.com	16,321	322.24	85.10	27,422.2	3,347.4
3	speedtest.wobcom.de	16,321	324.00	75.63	24,504.9	2,991.3
4	rychlost.poda.cz	15,636	361.34	38.76	14,005.5	1,709.7
5	spd-fisrv.hostkey.com	15,207	343.15	57.90	19,868.6	2,425.4
6	speed.itgate.net	16,292	346.52	33.27	11,529.8	1,407.4
7	porto.speedtest.net.zon.pt	17,761	318.44	13.77	4,383.9	535.2
8	speedtest.masnet.ec	13,621	353.72	14.23	5,034.5	614.6
9	dal.speedtest.clouvider.net	14,482	193.04	97.97	18,911.5	2,308.5
10	iperf.he.net	12,679	332.91	26.50	8,822.2	1,076.9
Average		15247.50	326.44	50.46	15,715.4	1,918.4

<Fig 4.3.1 Summary for mean RTT and mean Bandwidth>

Then we plot two bar charts with a linear scale and a logarithmic scale.



< Fig 4.3.2 Bar chart with a linear scale for 10 iperf servers>



< Fig 4.3.3 Bar chart with a logarithmic scale for 10 iperf servers>

Two scales are used here primarily because the minimum bandwidth-delay product (BDP) value for Host7 differs by nearly a factor of 7 from the maximum value for Host2, making a logarithmic scale entirely appropriate.

The Bandwidth-Delay Product (BDP) for each host is calculated using the formula $BDP = \text{average bandwidth} \times \text{average round-trip time (RTT)}$, where the average bandwidth values come from the iperf measurements in Section 4.1, and the average RTT values come from the ping measurements in Section 3.2.

Although `spd-desrv.hostkey.com` does not have the highest bandwidth, it has the highest BDP value (3,347 KB). This is because BDP is the product of bandwidth and latency: `spd-desrv.hostkey.com` achieves a high bandwidth of 85.1 Mbps, while its RTT is relatively high at 322.2ms, creating a large “pipeline” capable of accommodating a large volume of data transmission.[6] In contrast, `dal.speedtest.clouvider.net` recorded the highest measured bandwidth (98.0 Mbps) but the lowest round-trip time (RTT) (193.0ms), resulting in a BDP value of only 2,309 KB, ranking fifth. This illustrates an important characteristic of BDP: a low BDP value does not necessarily indicate poor network performance. `dal.speedtest.clouvider.net` provided the best throughput and the most stable connection among all 10 hosts, but its shorter round-trip time means that fewer bytes need to be transmitted at any given moment to keep the data transmission at full capacity. From a TCP performance perspective, a lower BDP value is actually easier to manage, as it requires a smaller TCP send window to achieve full link utilization [5].

The two lowest-ranked hosts, `porto.speedtest.net.zon.pt` (535 KB) and `speedtest.masnet.ec` (615 KB), both have BDP values well below the average of 1,918 KB. Their lower rankings are primarily due to their limited measured bandwidth (13.8 and 14.2 Mbps, respectively), rather than their RTT values. As described in Section 4.2, the Porto server appears to be constrained by a persistent low-capacity bottleneck near the destination, with a bandwidth under 13.8 Mbps, but the measured throughput of the Ecuador server is also low. Regardless of how high the RTT values are, BDP cannot be improved when available bandwidth remains so low.

The measurements were conducted on a shared Wi-Fi network, where observed download speeds were approximately 80 Mbps. This suggests that for hosts measuring throughput close to or exceeding this level, the local wireless environment may have influenced the measured throughput. At the same time, since some iperf test results exceeded the approximately 80 Mbps speed previously observed on the network, this value should be considered an approximate reference rather than a hard limit. Therefore, for hosts with the highest throughput, the true BDP may be slightly higher than the measured value, while for low-bandwidth hosts such as those in Porto and Ecuador, the primary bottleneck is more likely to occur on the remote path or at the destination.

Ans 4.4

Challenge 1: iperf.he.net refused default (sender) mode connections.

When running the standard iperf3 command (`iperf3 -c iperf.he.net`) against host 10 (iperf.he.net), the server rejected the connection in its default mode, where the client sends data to the server. This was not a network connectivity issue, but rather a server-side policy restriction. To resolve this issue, the `-R` flag was used (`iperf3 -c iperf.he.net -R`) [6], which reverses the direction of data flow so that the server sends data to the client. This allowed all three bandwidth measurements to succeed. Therefore, iperf3 was used for Host 10, while iperf2 was used for the remaining nine hosts.

Challenge 2: Significant jitter spike on the Ecuador server's path during the nighttime slot.

During the nighttime measurement in Section 4.2 (23:08 AEST), a traceroute to the Ecuadorian server showed significant fluctuations in the RTT at the 24th hop (voip.masnet.ec, 170.83.216.5), with the three probe results being 524.668ms, 327.077ms and 388.391ms respectively. In contrast, the RTT values recorded during a route trace of the same hop earlier in the morning were stable at about 327/327/328ms. The hostname of this hop contains the term “VoIP,” indicating that it likely belongs to the portion of the Ecuadorian ISP network associated with voice services. Therefore, the instability of this node during nighttime hours aligns with increased traffic load, but route tracing alone cannot prove that this node is the sole bottleneck. During the same period, the iperf bandwidth dropped to 20.8 Mbps compared with the other periods of the whole day. But RTT during this period didn't have a significant change.

In summary, these findings support the view that performance degradation was caused by changes in path conditions. At the same time, this shows that although bandwidth is affected by the line, RTT has not changed significantly.

References

- [1] db-ip.com, "IP Geolocation," db-ip.com. [Online]. Available: <https://db-ip.com/>. [Accessed: Apr. 2026].
- [2] R. W. Sinnott, "Virtues of the Haversine," *Sky and Telescope*, vol. 68, no. 2, p. 159, 1984.
- [3] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 8th ed. Harlow, U.K.: Pearson, 2021.
- [4] J. Postel, "Internet Control Message Protocol," RFC 792, Internet Engineering Task Force, Sep. 1981. [Online]. Available: <https://www.rfc-editor.org/rfc/rfc792>
- [5] V. Jacobson, R. Braden, and D. Borman, "TCP Extensions for High Performance," RFC 1323, Internet Engineering Task Force, May 1992. [Online]. Available: <https://www.rfc-editor.org/rfc/rfc1323>
- [6] ESnet, "iperf3: A TCP, UDP, and SCTP network bandwidth measurement tool," Energy Sciences Network, 2024. [Online]. Available: <https://software.es.net/iperf/>

Declaration

I acknowledge the use of Claude [claude.ai] to help me with the translation, grammar, and scientific terminology.

I use it to help me check for potential grammatical errors and issues with improper word usage in my writing.

Based on its output, I corrected the grammatical and wording errors in my writing. At the same time, I asked it to review the content against the assignment requirements to ensure I had met all the requirements.

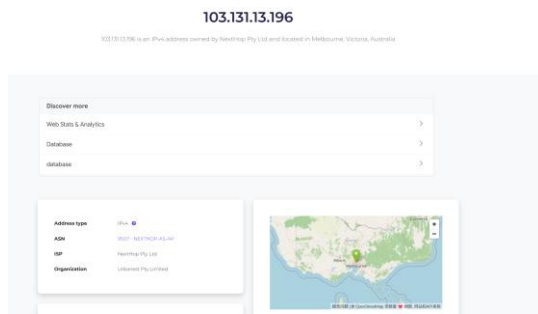
A full record of prompts and outputs is available upon request.

Appendix

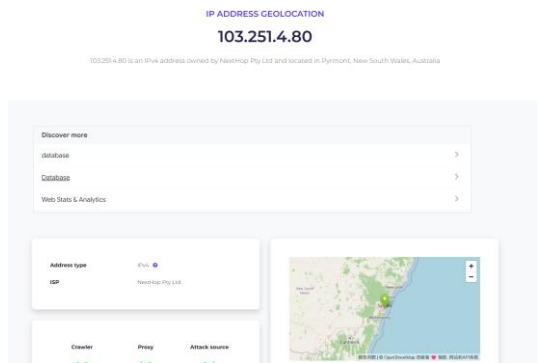
Section 2

```
rong@KevinRongdeMacBook-Air ~ % traceroute -q 1 -n -w 2 speedtest.awbb.net
traceroute to speedy.awbb.net (104.245.79.230), 64 hops max, 40 byte packets
 1 172.16.145.1 3.493 ms
 2 103.131.13.193 26.332 ms
 3 103.251.4.80 2.986 ms
 4 103.251.4.31 3.064 ms
 5 202.139.7.25 8.704 ms
 6 *
 7 *
 8 *
 9 61.88.33.48 18.847 ms
10 203.208.147.113 135.488 ms
11 203.208.154.14 141.910 ms
12 184.184.194.90 133.919 ms
13 203.208.166.61 136.772 ms
14 *
15 *
16 184.105.223.225 488.024 ms
17 184.184.197.243 283.966 ms
18 184.105.223.225 330.027 ms
19 104.245.79.3 322.437 ms
20 104.245.79.110 311.835 ms
21 104.245.79.3 389.575 ms
22 104.245.79.110 311.389 ms
23 104.245.79.230 405.384 ms !X
```

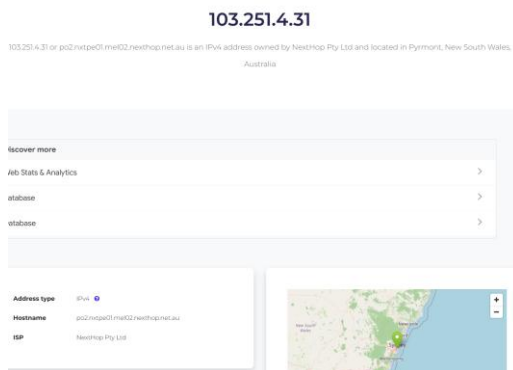
<Fig 2.1.2 traceroute command line for only one packet from my apartment to speedtest.awbb.net>



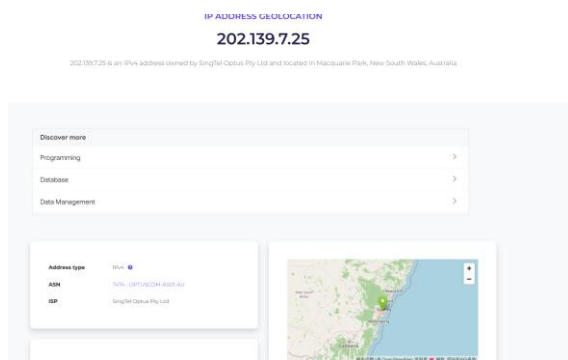
< Fig 2.1.3 IP of 2nd hop >



< Fig 2.1.4 IP of 3rd hop >



< Fig 2.1.5 IP of 4th hop >



< Fig 2.1.6 IP of 5th hop >

```

12 from math import radians, sin, cos, acos
13
14 R_EARTH_KM = 6371.0           # IUGG mean Earth radius
15 ORIGIN = (-37.8136, 144.9631)  # Melbourne CBD (privacy-preserving origin)
16
17
18 def great_circle_km(lat1: float, lon1: float, lat2: float, lon2: float) -> float:
19     """Return the great-circle distance in kilometres between two
20     (lat, lon) points, using the spherical law of cosines [Sinnott 1984]."""
21     phi1, phi2 = radians(lat1), radians(lat2)
22     dlam = radians(lon2 - lon1)
23     cos_c = min(1.0, sin(phi1) * sin(phi2) + cos(phi1) * cos(phi2) * cos(dlam))
24     return R_EARTH_KM * acos(cos_c)
25
26
27 # The 10 destinations, as looked up on db-ip.com (22 April 2026).
28 destinations = [
29     ("speedtest.awbb.net",      33.9845, -101.338),
30     ("spd-desrv.hostkey.com",   50.1109,  8.68213),
31     ("speedtest.wobcom.de",     50.1109,  8.68213),
32     ("rychlost.poda.cz",        49.8314, 18.2776),
33     ("spd-fisrv.hostkey.com",   60.1699, 24.9384),
34     ("speed.itgate.net",        45.4642,  9.18998),
35     ("porto.speedtest.net.zon.pt", 38.7599, -9.15765),
36     ("speedtest.masnet.ec",     -2.94017, -78.92),
37     ("dal.speedtest.clouvider.net", 32.7892, -96.8217),
38     ("iperf.he.net",            37.49,   -121.931),
39 ]
40
41
42 if __name__ == "__main__":
43     print(f"{'Hostname':<34} {'Lat':>9} {'Lon':>9} {'Dist (km)':>10}")
44     print("-" * 66)
45     for host, lat, lon in destinations:
46         d = great_circle_km(*ORIGIN, lat, lon)
47         print(f"{'host':<34} {'lat':>9.4f} {'lon':>9.4f} {'round(d):>10}")
48

```

<Fig 2.2.3 Python for distance>

```

ron@KevinRongdeMacBook-Air ~ % traceroute -n -w 2 -m 30 -q 4 speedtest.awbb.net
traceroute to speedtest.awbb.net (184.245.79.230), 30 hops max, 40 byte packets
 1 172.16.145.1 3.415 ms 3.409 ms 3.267 ms 3.293 ms
 2 103.131.13.193 25.963 ms 21.800 ms 22.060 ms 21.318 ms
 3 183.251.4.89 2.970 ms 3.470 ms 3.458 ms 3.472 ms
 4 183.251.4.31 3.229 ms 3.733 ms 2.898 ms 2.678 ms
 5 202.139.7.25 4.334 ms 4.161 ms 4.297 ms 4.315 ms
 6 * * * *
 7 * * * *
 8 * 59.154.142.63 55.935 ms *
 9 * * * *
10 203.208.147.65 136.286 ms 136.006 ms *
11 203.208.147.113 159.283 ms
12 61.88.33.48 17.799 ms 18.289 ms 18.321 ms 19.386 ms
13 184.184.194.90 133.666 ms 133.858 ms 134.218 ms 134.418 ms
14 184.184.194.90 134.318 ms 132.787 ms
15 184.184.199.145 178.858 ms 177.748 ms
16 * 184.184.188.97 179.559 ms
17 184.184.197.243 299.043 ms *
18 184.184.199.245 297.976 ms
19 184.184.199.245 297.976 ms
20 184.184.199.145 178.185 ms
21 184.184.197.243 329.302 ms
22 184.184.197.243 348.266 ms * 350.817 ms
23 184.184.223.225 307.492 ms 306.831 ms
24 184.184.226.98 409.703 ms *
25 184.184.197.243 349.934 ms
26 184.245.79.3 489.364 ms
27 184.184.197.243 307.488 ms *
28 184.245.79.118 349.798 ms

```

```

9 * * * *
10 203.208.147.65 136.286 ms 136.006 ms *
11 203.208.147.113 159.283 ms
12 61.88.33.48 17.799 ms 18.289 ms 18.321 ms 19.386 ms
13 * 203.208.166.61 142.021 ms *
14 184.184.194.90 134.318 ms 132.787 ms
15 * 184.184.188.97 179.559 ms
16 184.184.197.243 299.043 ms *
17 * 184.184.199.245 297.976 ms
18 184.184.223.225 307.492 ms 306.831 ms
19 184.184.197.243 349.934 ms
20 184.184.197.243 307.688 ms *
21 184.245.79.118 349.798 ms
22 184.184.226.98 409.281 ms 384.477 ms
23 184.245.79.118 369.662 ms
24 184.245.79.3 409.706 ms
25 184.245.79.230 409.935 ms IX
26 184.245.79.3 311.423 ms 404.754 ms
27 184.245.79.118 410.189 ms 408.927 ms 410.332 ms 410.843 ms
28 184.245.79.230 316.725 ms IX 318.717 ms IX
ron@KevinRongdeMacBook-Air ~ %

```

<Fig 2.2.4(a) speedtest.awbb.net>

< Fig 2.2.4(b) speedtest.awbb.net>

Hostname	speedtest.awbb.net
ASN	AS228 - AMERICA ONLINE
ISP	America Online
Connection	Wireless

Estimated threat level for this IP address is **LOW**

Country	United States
State / Region	Texas
District / County	El Paso
City	El Paso
Zip / Postal code	79901
Weather station	KCT0640 - El Paso
Coordinates	31.7895, -106.388
Timezone	America/Chicago (UTC-5)
Local time	2024-03-15

<Fig 2.2.4(c) speedtest.awbb.net>

```

rong@KevinRongdeMacBook-Air ~ % sudo traceroute -I -q 4 -n -w 2 -m 30 spd-desrv.hostkey.com
Password:
traceroute to spd-desrv.hostkey.com (185.84.224.68), 30 hops max, 48 byte packets
 1 172.16.145.1 3.889 ms 3.897 ms 3.235 ms 3.177 ms
 2 183.131.13.193 14.904 ms 15.261 ms 18.044 ms 14.715 ms
 3 183.251.4.80 2.762 ms 3.126 ms 3.258 ms 3.201 ms
 4 183.251.4.31 3.224 ms 3.253 ms 3.260 ms 3.298 ms
 5 202.139.7.25 4.624 ms 4.210 ms 4.211 ms 4.204 ms
 6 * * * *
 7 * * * *
 8 * * * *
 9 * * * *
10 59.154.57.154 36.792 ms 32.023 ms * 34.893 ms
11 61.88.33.48 18.142 ms 18.111 ms 18.263 ms 17.938 ms
12 203.208.147.65 147.991 ms 136.136 ms 137.735 ms 135.740 ms
13 203.208.166.61 137.729 ms 136.537 ms 136.762 ms 145.006 ms
14 184.184.194.90 134.068 ms 134.740 ms 134.382 ms 133.746 ms
15 * * 184.184.199.181 134.530 ms 137.689 ms
16 * * * *
17 * * * *
18 * 184.184.189.52 315.439 ms *
19 216.66.88.175 281.551 ms 301.694 ms 283.308 ms 281.246 ms
20 185.84.224.68 371.063 ms 279.666 ms 334.481 ms 307.267 ms
rong@KevinRongdeMacBook-Air ~ %

```

<Fig2.2.5(a) spd-desrv.hostkey.com>

ASN

57043 - HOSTKEY-AS

ISP

Hostkey B.V.

Connection

I-Hosting

Organization

Hostkey B.V.

Crawler

Proxy

Attack source

Estimated threat level for this IP address is

LOW

Country

Germany

State / Region

Hesse
黑森州

District / County

Regierungsbezirk Darmstadt

City

Frankfurt am Main
法兰克福

Zip / Postal code

60313

Weather station

CM000040 - Frankfurt am Main

Coordinates

50.109, 8.68213

Timezone

Europe/Berlin (UTC+2)

Local time

07:44:23

<Fig2.2.5(b) spd-desrv.hostkey.com>

```

rong@KevinRongdeMacBook-Air ~ % sudo traceroute -I -q 4 -n -w 2 -m 30 speedtest.wobcom.de
Password:
traceroute to speedtest.wobcom.de (62.176.224.217), 30 hops max, 48 byte packets
 1 172.16.145.1 3.833 ms 3.159 ms 3.500 ms 3.190 ms
 2 183.131.13.193 9.604 ms 22.089 ms 11.225 ms 21.828 ms
 3 183.251.4.80 2.970 ms 3.407 ms 3.447 ms 3.159 ms
 4 183.251.4.31 3.181 ms 3.490 ms 3.266 ms 3.714 ms
 5 202.139.7.25 3.494 ms 4.424 ms 4.641 ms 4.463 ms
 6 * * * *
 7 * * * *
 8 * * * *
 9 * * * *
10 59.154.57.198 44.504 ms * * *
11 61.88.33.48 22.365 ms 18.993 ms 19.082 ms 19.251 ms
12 203.208.147.65 136.767 ms 136.616 ms 137.983 ms 137.453 ms
13 80.156.163.2 322.768 ms 306.427 ms 237.970 ms 238.121 ms
14 217.239.39.38 344.984 ms 331.088 ms 330.676 ms 362.195 ms
15 80.157.200.207 407.942 ms 420.236 ms 398.609 ms 409.508 ms
16 62.176.224.217 303.740 ms 304.815 ms 303.533 ms 316.751 ms
rong@KevinRongdeMacBook-Air ~ %

```

<Fig 2.2.6(a) speedtest.wobcom.de>

ASN

9136 - WOBCOM

ISP

Wobcom GmbH

Organization

Wobcom

Crawler

Proxy

Attack source

Estimated threat level for this IP address is

LOW

Country

Germany

State / Region

Hesse
黑森州

District / County

Regierungsbezirk Darmstadt

City

Frankfurt am Main
法兰克福

Zip / Postal code

60313

Weather station

CM000040 - Frankfurt am Main

Coordinates

50.109, 8.68213

Timezone

Europe/Berlin (UTC+2)

Local time

07:44:30

Languages

de

<Fig 2.2.6(b) speedtest.wobcom.de>


```
30 * * * *
rong@KevinRongdeMacBook-Air ~ % traceroute -n -w 2 -m 30 -q 4 rychlost.poda.cz
traceroute to rychlost.poda.cz (86.135.13.2), 30 hops max, 48 byte packets
 1 172.16.145.1 3.257 ms 2.788 ms 3.466 ms 3.184 ms
 2 183.131.13.193 244.199 ms 23.313 ms 17.434 ms 17.070 ms
 3 183.251.4.80 2.862 ms 3.192 ms 3.196 ms 3.211 ms
 4 183.251.4.31 3.130 ms 3.216 ms 3.334 ms 2.977 ms
 5 138.217.286.217 3.121 ms 3.515 ms 3.622 ms 3.173 ms
 6 203.50.11.10 5.261 ms 4.127 ms 4.469 ms 3.197 ms
 7 203.50.13.124 5.788 ms 4.669 ms 5.868 ms 5.977 ms
 8 203.50.13.132 16.265 ms 16.274 ms 15.880 ms 14.693 ms
 9 203.50.6.186 16.567 ms 16.780 ms 16.882 ms 16.287 ms
10 203.50.13.94 14.946 ms 16.915 ms 17.187 ms 14.898 ms
11 203.50.13.94 153.269 ms 151.883 ms 151.514 ms 153.144 ms
12 202.84.141.225 153.273 ms 152.829 ms 152.597 ms 152.496 ms
13 202.84.141.225 153.826 ms 151.639 ms 154.638 ms 151.838 ms
14 202.84.253.86 150.917 ms 151.565 ms 151.546 ms 151.191 ms
15 62.115.142.134 143.268 ms 161.379 ms 144.880 ms 143.744 ms
16 62.115.148.156 161.322 ms * 165.208 ms 161.402 ms
17 62.115.148.157 387.496 ms
18 62.115.148.247 386.944 ms 387.387 ms
19 62.115.121.221 387.643 ms
20 62.115.116.213 386.641 ms
21 62.115.121.221 228.858 ms 291.896 ms
22 62.115.134.236 288.212 ms
23 62.115.148.186 283.382 ms
24 62.115.143.236 324.833 ms *
25 62.115.122.139 358.347 ms 281.783 ms
26 62.115.127.132 221.224 ms 228.882 ms
27 62.115.124.29 388.921 ms
28 62.115.136.280 293.874 ms
29 62.115.124.29 288.380 ms 286.468 ms
30 62.115.122.139 282.882 ms
31 62.115.38.66 367.886 ms
32 62.115.122.139 283.380 ms 387.142 ms
33 86.135.13.2 388.453 ms IZ
34 62.115.124.29 289.616 ms
35 86.135.13.2 323.441 ms IZ 418.135 ms IZ
rong@KevinRongdeMacBook-Air ~ %
```

< Fig 2.2.7(a) rychlost.poda.cz >

< Fig 2.2.7(b) rychlost.poda.cz >

Hostname rychlost-bb.poda.cz

ASN 30764 - PODA-AS

ISP PODA a.s.

Organization Podia

Crawler Proxy Attack source

Estimated threat level for this IP address is **LOW**



Country Czechia

State / Region Moravskoslezský

District / County Okres Ostrava-město

City Ostrava (Moravská Ostrava a Přívoz)

Zip / Postal code 702 00

Weather station EZXX0009 - Ostrava

Coordinates 49.8314, 18.2776

Timezone Europe/Prague (UTC+2)

Local time 07:46:03

Languages cs, sk

<Fig 2.2.7(c) rychlost.poda.cz >

```
rong@KevinRongdeMacBook-Air ~ % sudo traceroute -I -q 4 -n -w 2 -m 30 spd-fisrv.hostkey.com
traceroute to spd-fisrv.hostkey.com (212.237.217.41), 30 hops max, 48 byte packets
 1 172.16.145.1 3.814 ms 3.170 ms 3.183 ms 3.179 ms
 2 183.131.13.193 15.640 ms 19.424 ms 21.339 ms 17.117 ms
 3 183.251.4.80 3.541 ms 2.689 ms 3.257 ms 3.175 ms
 4 183.251.4.31 3.263 ms 3.383 ms 3.490 ms 3.168 ms
 5 138.217.286.217 3.696 ms 3.156 ms 3.146 ms 3.195 ms
 6 203.50.11.10 5.286 ms 5.437 ms 4.150 ms 4.230 ms
 7 203.50.13.124 4.250 ms 5.341 ms 4.284 ms 4.259 ms
 8 203.50.13.132 16.277 ms 14.344 ms 13.651 ms 15.148 ms
 9 203.50.6.186 14.943 ms 14.712 ms 16.118 ms 15.951 ms
10 203.50.13.94 14.799 ms 15.366 ms 14.785 ms 15.888 ms
11 203.50.13.94 153.358 ms 151.686 ms 153.081 ms 153.550 ms
12 202.84.141.225 153.537 ms 152.231 ms 151.645 ms 152.572 ms
13 202.84.141.225 153.033 ms 152.681 ms 152.477 ms 152.370 ms
14 202.84.253.86 154.788 ms 153.634 ms 151.358 ms 151.507 ms
15 218.57.53.99 161.823 ms 161.538 ms 165.739 ms 160.883 ms
16 141.136.111.114 290.133 ms 395.277 ms 387.299 ms 290.189 ms
17 195.72.86.226 267.411 ms 363.573 ms 389.014 ms 268.371 ms
18 * * * *
19 212.237.217.41 315.521 ms 293.554 ms 293.935 ms 293.997 ms
rong@KevinRongdeMacBook-Air ~ %
```

< Fig 2.2.8(a) spd-fisrv.hostkey.com >

ASN 57043 - HOSTKEY-AS

ISP Hostkey B.V.

Connection Hosting

Organization Hostkey B.V.

Crawler Proxy Attack source

Estimated threat level for this IP address is **LOW**



Country Finland

State / Region Uusimaa

District / County Helsinki sub-region

City Helsinki

Zip / Postal code 00121

Weather station FBOX002 - Helsinki

Coordinates 60.1693, 24.9384

Timezone Europe/Helsinki (UTC+3)

Local time 08:46:54

< Fig 2.2.8(b) spd-fisrv.hostkey.com >

```

rong@KevinRongdeMacBook-Air ~ % sudo traceroute -I -q 4 -n -w 2 -m 30 speed.itgate.net
traceroute to speed.itgate.net (213.212.129.36), 30 hops max, 48 byte packets
 1 172.16.145.1 4.178 ms 3.283 ms 3.429 ms 3.367 ms
 2 103.131.13.193 15.140 ms 17.141 ms 15.074 ms 21.596 ms
 3 103.251.4.80 3.460 ms 2.840 ms 3.487 ms 3.162 ms
 4 103.251.4.31 3.169 ms 3.169 ms 3.193 ms 3.231 ms
 5 138.217.205.217 2.670 ms 3.271 ms 3.223 ms 3.149 ms
 6 203.50.11.10 5.306 ms 4.666 ms 4.215 ms 3.477 ms
 7 203.50.13.124 6.328 ms 6.078 ms 5.962 ms 5.251 ms
 8 203.50.13.132 16.459 ms 15.666 ms 15.979 ms 14.969 ms
 9 203.50.6.106 14.853 ms 17.146 ms 15.928 ms 15.936 ms
10 203.50.13.94 16.294 ms 15.966 ms 17.026 ms 15.342 ms
11 203.50.13.94 153.533 ms 152.901 ms 153.356 ms 152.495 ms
12 202.84.141.225 151.796 ms 152.849 ms 152.846 ms 151.119 ms
13 202.84.141.225 152.138 ms 152.794 ms 152.359 ms 151.744 ms
14 202.84.253.06 151.502 ms 151.074 ms 150.806 ms 151.143 ms
15 210.67.53.99 161.707 ms 162.151 ms 161.302 ms 165.381 ms
16 89.149.142.254 388.521 ms 302.569 ms 302.782 ms 315.844 ms
17 46.33.86.66 409.649 ms 307.407 ms 408.657 ms 409.872 ms
18 213.254.30.195 512.157 ms 305.968 ms 305.962 ms 306.104 ms
19 213.254.30.180 305.608 ms 311.831 ms 308.701 ms 305.734 ms
20 * * * *
21 * * * *
22 213.212.129.36 311.843 ms 307.886 ms 342.774 ms 409.469 ms
rong@KevinRongdeMacBook-Air ~ %

```

< Fig 2.2.9(a) speed.itgate.net >

Address type	IPv4
Hostname	speed.itgate.net
ASN	12779 - ITGATE
ISP	ITGate S.p.A.
Organization	ITGate S.p.A.

Crawler Proxy Attack source

Estimated threat level for this IP address is **LOW**

Country: Italy

State / Region: Lombardy

District / County: Province of Milan

City: Milan

Zip / Postal code: 2012

Weather station: TX00042 - Milan

Coordinates: 45.4642, 9.18998

Timezone: Europe/Rome (UTC+2)

Local time: 07:48:16

Languages: it-IT, de-IT, fr-IT, sc, ca, co, sl

< Fig 2.2.9(b) speed.itgate.net >

```

rong@KevinRongdeMacBook-Air ~ % sudo traceroute -I -q 4 -n -w 2 -m 30 porto.speedtest.net.zon.pt
traceroute to porto.speedtest.net.zon.pt (95.95.252.95), 30 hops max, 48 byte packets
 1 172.16.145.1 4.660 ms 3.129 ms 3.001 ms 2.473 ms
 2 103.131.13.193 20.256 ms 25.805 ms 14.803 ms 22.569 ms
 3 103.251.4.80 2.820 ms 3.182 ms 3.221 ms 3.177 ms
 4 103.251.4.31 3.136 ms 3.826 ms 2.586 ms 3.308 ms
 5 202.139.7.25 4.331 ms 4.089 ms 4.324 ms 4.257 ms
 6 * * * *
 7 * * * *
 8 172.26.11.185 54.130 ms 52.115 ms 52.761 ms 51.749 ms
 9 * * * *
10 154.54.39.225 266.029 ms 250.886 ms 260.826 ms 249.792 ms
11 154.54.61.102 262.424 ms 262.781 ms 350.942 ms 262.262 ms
12 154.54.61.213 351.989 ms 262.098 ms 261.649 ms 261.373 ms
13 * * * *
14 * * * *
15 95.95.252.95 275.735 ms 306.638 ms 307.037 ms 267.664 ms
rong@KevinRongdeMacBook-Air ~ %

```

< Fig 2.2.10(a) porto.speedtest.net.zon.pt >

Address type	IPv4
Hostname	porto.speedtest.net.zon.pt
ASN	2860 - NOS COMUNICACOES
ISP	Nos Comunicacoes S.A.
Organization	Nos Comunicacoes S.A.

Crawler Proxy Attack source

Estimated threat level for this IP address is **LOW**

Country: Portugal

State / Region: Lisbon

City: Lisbon

Weather station: POXXX016 - Lisbon

Coordinates: 38.7599, -9.15765

Timezone: Europe/Lisbon (UTC+1)

Local time: 06:49:09

Languages: pt-PT, mwl

Currency: Euro (EUR)

< Fig 2.2.10(b) porto.speedtest.net.zon.pt >

```

rong@KevinRongdeMacBook-Air ~ % traceroute -n -w 2 -m 30 -q 4 speedtest.masnet.ec
traceroute to speedtest.masnet.ec (170.83.216.19), 30 hops max, 40 byte packets
 1 172.16.145.1 3.993 ms 3.191 ms 3.170 ms 3.170 ms
 2 103.131.13.193 20.748 ms 21.560 ms 15.773 ms 22.601 ms
 3 103.251.4.80 2.998 ms 3.168 ms 3.141 ms 3.055 ms
 4 103.251.4.31 3.200 ms 3.234 ms 3.188 ms 3.208 ms
 5 202.139.7.25 4.453 ms 4.213 ms 4.286 ms 4.303 ms
 6 * * * * *
 7 * * * * *
 8 172.26.11.185 52.706 ms 52.013 ms 52.252 ms 52.354 ms
 9 * * * * *
10 154.54.42.181 225.112 ms
   154.54.164.289 202.839 ms
   154.54.25.149 202.107 ms
   154.54.42.181 254.875 ms
11 154.54.45.161 221.366 ms
   154.54.42.181 214.012 ms
   154.54.45.161 279.538 ms
   154.54.44.85 309.978 ms
12 154.54.166.57 266.142 ms
   154.54.44.85 242.859 ms
   154.54.166.69 307.354 ms 306.982 ms
13 154.54.165.25 266.510 ms
   154.54.166.57 245.939 ms
   154.54.165.25 309.521 ms
   154.54.165.29 307.244 ms
14 154.54.40.250 509.808 ms
   154.54.41.54 613.170 ms
   154.54.165.25 307.770 ms
   154.54.41.54 389.128 ms
15 * 154.54.5.162 370.374 ms 306.605 ms 307.608 ms
16 154.54.168.85 307.231 ms 274.982 ms 339.694 ms 306.432 ms
17 38.104.94.178 284.441 ms
   154.54.40.250 329.422 ms
   38.104.94.178 309.299 ms
   38.88.57.26 280.606 ms
18 69.79.100.25 332.058 ms 307.700 ms
   38.88.57.26 284.709 ms
   69.79.100.25 272.025 ms
19 69.79.100.135 364.601 ms * * *
20 * 190.242.173.79 312.240 ms * *
21 * * 190.242.173.79 315.475 ms *
22 * * * * *
23 * * 190.242.173.79 426.382 ms *
24 170.83.216.5 350.325 ms 408.047 ms * 350.041 ms
25 170.83.216.5 410.342 ms 411.011 ms * 347.756 ms
26 170.83.216.19 409.549 ms 512.055 ms 409.365 ms 411.807 ms
rong@KevinRongdeMacBook-Air ~ %

```

< Fig 2.2.11(a) speedtest.masnet.ec >

Address type IPv4

Hostname speedtest.masnet.ec

ASN 254851 - AS264851

ISP ifotoncorp S.A.



Crawler X

Proxy X

Attack source X

Estimated threat level for this IP address is LOW

Country Ecuador 🇪🇨

State / Region Azuay
阿苏艾省

District / County Cuenca

City Santa Ana

Zip / Postal code 010215

Weather station E0X0157 - Santa Ana

Coordinates -2.94017, -78.92

Timezone America/Guayaquil (UTC-5)

< Fig 2.2.11(b) speedtest.masnet.ec >

```

rong@KevinRongdeMacBook-Air ~ % traceroute -n -w 2 -m 30 -q 4 dal.speedtest.clouvider.net
traceroute to dal.speedtest.clouvider.net (2.56.188.136), 30 hops max, 40 byte packets
 1 172.16.145.1 8.382 ms 2.610 ms 3.586 ms 3.252 ms
 2 103.131.13.193 16.266 ms 22.338 ms 16.181 ms 21.550 ms
 3 103.251.4.80 3.093 ms 3.150 ms 3.281 ms 3.216 ms
 4 103.251.4.31 3.255 ms 3.153 ms 3.136 ms 3.195 ms
 5 138.217.205.217 3.611 ms 3.788 ms 3.984 ms 3.086 ms
 6 203.50.11.10 3.918 ms 4.211 ms 5.511 ms 5.123 ms
 7 203.50.13.124 5.231 ms 4.213 ms 6.376 ms 3.947 ms
 8 203.50.13.132 16.483 ms 21.249 ms 22.951 ms 15.757 ms
 9 203.50.6.106 55.201 ms 15.715 ms 14.838 ms 16.239 ms
10 203.50.13.94 14.937 ms 14.821 ms 16.098 ms 14.901 ms
11 203.50.13.94 150.924 ms 151.377 ms 151.712 ms 153.251 ms
12 202.84.141.225 151.774 ms 152.462 ms 153.176 ms 152.999 ms
13 202.84.141.225 153.679 ms 153.947 ms 151.671 ms 152.372 ms
14 202.84.253.86 151.498 ms 151.336 ms 150.791 ms 153.932 ms
15 210.57.53.99 161.429 ms 161.554 ms 162.032 ms 162.201 ms
16 213.200.115.26 179.476 ms 180.491 ms 179.501 ms 179.700 ms
17 76.74.1.18 190.807 ms 190.795 ms 191.534 ms 191.213 ms
18 2.56.188.1 188.662 ms 201.073 ms 182.869 ms 184.575 ms
19 2.56.188.136 191.329 ms 191.571 ms 191.264 ms 191.032 ms
rong@KevinRongdeMacBook-Air ~ %

```

< Fig 2.2.12(a) dal.speedtest.clouvider.net >

Address type

IPv4

ASN

62240 - Clouvider

ISP

Clouvider Limited

Connection

Hosting

Crawler

×

Proxy

×

Attack source

×

Estimated threat level for this IP address is

LOW

Country

United States

State / Region

Texas

City

Dallas

Zip / Postal code

75207

Weather station

USTX0327 - Dallas

Coordinates

32.7892, -96.8217

Timezone

America/Chicago (UTC-5)

Local time

00:50:29

< Fig 2.2.12(b) dal.speedtest.clouvider.net >

```

rong@KevinRongdeMacBook-Air ~ % traceroute -q 4 -n -w 2 -m 30 iperf.he.net
traceroute to 1580.mtu.he.net (216.218.207.42), 30 hops max, 40 byte packets
 1 172.16.145.1 3.783 ms 3.169 ms 3.164 ms 3.173 ms
 2 183.131.13.193 21.606 ms 18.581 ms 24.741 ms 36.462 ms
 3 183.251.4.80 2.788 ms 3.280 ms 3.075 ms 3.387 ms
 4 183.251.4.31 3.226 ms 2.982 ms 3.615 ms 3.031 ms
 5 282.139.7.25 4.231 ms 4.287 ms 3.284 ms 4.435 ms
 6 * * * *
 7 * * * *
 8 * * * *
 9 * * * *
10 61.88.33.47 34.531 ms 28.887 ms 29.195 ms 29.184 ms
11 283.288.158.189 136.623 ms
   283.288.158.173 137.377 ms 139.508 ms
   283.288.158.189 135.932 ms
12 283.288.158.197 137.051 ms
   283.288.158.193 137.674 ms
   283.288.178.13 137.983 ms
   283.288.178.17 139.325 ms
13 283.288.166.61 136.133 ms 136.655 ms 137.443 ms 137.664 ms
14 184.184.194.90 134.755 ms 135.021 ms 134.943 ms 134.430 ms
15 * * * 184.184.199.181 138.627 ms
16 * * * 184.184.199.10 179.658 ms *
17 * * * 184.184.194.37 385.956 ms
18 72.52.92.245 387.842 ms 284.674 ms * 374.981 ms
19 184.184.197.203 290.131 ms 324.665 ms 285.837 ms 327.899 ms
20 184.184.189.166 307.231 ms 310.990 ms 303.984 ms 291.497 ms
21 216.218.207.42 285.575 ms 284.137 ms 366.898 ms 307.151 ms
rong@KevinRongdeMacBook-Air ~ %

```

< Fig 2.2.13(a) iperf.he.net >

ASN

6939 - HURRICANE

ISP

Hurricane Electric LLC

Connection

Hosting

Organization

Hurricane Electric

Crawler

×

Proxy

×

Attack source

×

Estimated threat level for this IP address is

LOW

Country

United States

State / Region

California

District / County

Alameda

City

Fremont (East Industrial)

Zip / Postal code

94539

Weather station

USCA0403 - Fremont

Coordinates

37.49, -121.931

Timezone

America/Los_Angeles (UTC-7)

Local time

00:31:08

< Fig 2.2.13(b) iperf.he.net >

Section 3


```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.awbb.net
PING speedy.awbb.net (104.245.79.230): 56 data bytes
64 bytes from 104.245.79.230: icmp_seq=0 ttl=48 time=310.000 ms
64 bytes from 104.245.79.230: icmp_seq=1 ttl=48 time=314.050 ms
64 bytes from 104.245.79.230: icmp_seq=2 ttl=48 time=314.471 ms
64 bytes from 104.245.79.230: icmp_seq=3 ttl=48 time=392.554 ms

--- speedy.awbb.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 310.000/332.769/392.554/34.561 ms
rong@KevinRongdeMacBook-Air ~ %

```

<Fig 3.1.2 ping for speedtest.awbb.net>

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.awbb.net
PING speedy.awbb.net (104.245.79.230): 56 data bytes
64 bytes from 104.245.79.230: icmp_seq=0 ttl=48 time=323.385 ms
64 bytes from 104.245.79.230: icmp_seq=1 ttl=48 time=344.044 ms
64 bytes from 104.245.79.230: icmp_seq=2 ttl=48 time=469.762 ms
64 bytes from 104.245.79.230: icmp_seq=3 ttl=48 time=489.509 ms

--- speedy.awbb.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 323.385/406.673/489.509/73.655 ms
rong@KevinRongdeMacBook-Air ~ %

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.awbb.net
PING speedy.awbb.net (104.245.79.230): 56 data bytes
64 bytes from 104.245.79.230: icmp_seq=0 ttl=48 time=310.689 ms
64 bytes from 104.245.79.230: icmp_seq=1 ttl=48 time=333.769 ms
64 bytes from 104.245.79.230: icmp_seq=2 ttl=48 time=352.317 ms
64 bytes from 104.245.79.230: icmp_seq=3 ttl=48 time=373.521 ms

--- speedy.awbb.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 310.689/342.574/373.521/23.167 ms

```

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.awbb.net
PING speedy.awbb.net (104.245.79.230): 56 data bytes
64 bytes from 104.245.79.230: icmp_seq=0 ttl=48 time=401.844 ms
64 bytes from 104.245.79.230: icmp_seq=1 ttl=48 time=323.581 ms
64 bytes from 104.245.79.230: icmp_seq=2 ttl=48 time=342.975 ms
64 bytes from 104.245.79.230: icmp_seq=3 ttl=48 time=362.954 ms

--- speedy.awbb.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 323.581/357.839/401.844/28.970 ms
rong@KevinRongdeMacBook-Air ~ %

```

<Fig 3.2.1 3 times ping for speedtest.awbb.net>

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 spd-desrv.hostkey.com
PING spd-desrv.hostkey.com (185.84.224.68): 56 data bytes
64 bytes from 185.84.224.68: icmp_seq=0 ttl=52 time=349.416 ms
64 bytes from 185.84.224.68: icmp_seq=1 ttl=52 time=280.900 ms
64 bytes from 185.84.224.68: icmp_seq=2 ttl=52 time=389.726 ms
64 bytes from 185.84.224.68: icmp_seq=3 ttl=52 time=387.432 ms

--- spd-desrv.hostkey.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 280.900/331.869/389.726/41.383 ms
rong@KevinRongdeMacBook-Air ~ %

rong@KevinRongdeMacBook-Air ~ % ping -c 4 spd-desrv.hostkey.com
PING spd-desrv.hostkey.com (185.84.224.68): 56 data bytes
64 bytes from 185.84.224.68: icmp_seq=0 ttl=52 time=313.697 ms
64 bytes from 185.84.224.68: icmp_seq=1 ttl=52 time=332.636 ms
64 bytes from 185.84.224.68: icmp_seq=2 ttl=52 time=351.454 ms
64 bytes from 185.84.224.68: icmp_seq=3 ttl=52 time=370.486 ms

--- spd-desrv.hostkey.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 313.697/342.068/370.486/21.152 ms
rong@KevinRongdeMacBook-Air ~ %

rong@KevinRongdeMacBook-Air ~ % ping -c 4 spd-desrv.hostkey.com
PING spd-desrv.hostkey.com (185.84.224.68): 56 data bytes
64 bytes from 185.84.224.68: icmp_seq=0 ttl=52 time=284.299 ms
64 bytes from 185.84.224.68: icmp_seq=1 ttl=52 time=291.784 ms
64 bytes from 185.84.224.68: icmp_seq=2 ttl=52 time=310.441 ms
64 bytes from 185.84.224.68: icmp_seq=3 ttl=52 time=284.299 ms

--- spd-desrv.hostkey.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 284.299/292.768/310.441/18.637 ms
rong@KevinRongdeMacBook-Air ~ %

```

<Fig 3.2.2 3 times ping for spd-desrv.hostkey.com>

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.wobcom.de
PING speedtest.wobcom.de (62.176.224.217): 56 data bytes
64 bytes from 62.176.224.217: icmp_seq=0 ttl=50 time=303.360 ms
64 bytes from 62.176.224.217: icmp_seq=1 ttl=50 time=307.364 ms
64 bytes from 62.176.224.217: icmp_seq=2 ttl=50 time=345.282 ms
64 bytes from 62.176.224.217: icmp_seq=3 ttl=50 time=363.234 ms

--- speedtest.wobcom.de ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 303.360/329.810/363.234/25.298 ms
rong@KevinRongdeMacBook-Air ~ %

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.wobcom.de
PING speedtest.wobcom.de (62.176.224.217): 56 data bytes
64 bytes from 62.176.224.217: icmp_seq=0 ttl=50 time=302.931 ms
64 bytes from 62.176.224.217: icmp_seq=1 ttl=50 time=379.141 ms
64 bytes from 62.176.224.217: icmp_seq=2 ttl=50 time=307.464 ms
64 bytes from 62.176.224.217: icmp_seq=3 ttl=50 time=315.072 ms

--- speedtest.wobcom.de ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 302.931/326.152/379.141/30.899 ms
rong@KevinRongdeMacBook-Air ~ %

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.wobcom.de
PING speedtest.wobcom.de (62.176.224.217): 56 data bytes
64 bytes from 62.176.224.217: icmp_seq=0 ttl=50 time=303.261 ms
64 bytes from 62.176.224.217: icmp_seq=1 ttl=50 time=320.024 ms
64 bytes from 62.176.224.217: icmp_seq=2 ttl=50 time=338.643 ms
64 bytes from 62.176.224.217: icmp_seq=3 ttl=50 time=302.182 ms

--- speedtest.wobcom.de ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 302.182/316.028/338.643/14.850 ms

```

<Fig 3.2.3 3 times ping for speedtest.wobcom.de>

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 rychlost.poda.cz
PING rychlost.poda.cz (85.135.13.2): 56 data bytes
64 bytes from 85.135.13.2: icmp_seq=0 ttl=36 time=325.757 ms
64 bytes from 85.135.13.2: icmp_seq=1 ttl=36 time=345.009 ms
64 bytes from 85.135.13.2: icmp_seq=2 ttl=36 time=364.614 ms
64 bytes from 85.135.13.2: icmp_seq=3 ttl=36 time=383.288 ms

--- rychlost.poda.cz ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 325.757/354.667/383.288/21.489 ms
rong@KevinRongdeMacBook-Air ~ %

```

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 rychlost.poda.cz
PING rychlost.poda.cz (85.135.13.2): 56 data bytes
64 bytes from 85.135.13.2: icmp_seq=0 ttl=36 time=336.553 ms
64 bytes from 85.135.13.2: icmp_seq=1 ttl=36 time=355.891 ms
64 bytes from 85.135.13.2: icmp_seq=2 ttl=36 time=380.725 ms
64 bytes from 85.135.13.2: icmp_seq=3 ttl=36 time=398.892 ms

--- rychlost.poda.cz ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 336.553/368.015/398.892/23.726 ms
rong@KevinRongdeMacBook-Air ~ %

```

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 rychlost.poda.cz
PING rychlost.poda.cz (85.135.13.2): 56 data bytes
64 bytes from 85.135.13.2: icmp_seq=0 ttl=36 time=359.989 ms
64 bytes from 85.135.13.2: icmp_seq=1 ttl=36 time=378.885 ms
64 bytes from 85.135.13.2: icmp_seq=2 ttl=36 time=397.973 ms
64 bytes from 85.135.13.2: icmp_seq=3 ttl=36 time=389.479 ms

--- rychlost.poda.cz ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 389.479/361.332/397.973/32.958 ms
rong@KevinRongdeMacBook-Air ~ %

```

<Fig 3.2.4 3 times ping for rychlost.poda.cz>

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 spd-fisrv.hostkey.com
PING spd-fisrv.hostkey.com (212.237.217.41): 56 data bytes
64 bytes from 212.237.217.41: icmp_seq=0 ttl=54 time=395.802 ms
64 bytes from 212.237.217.41: icmp_seq=1 ttl=54 time=322.285 ms
64 bytes from 212.237.217.41: icmp_seq=2 ttl=54 time=294.590 ms
64 bytes from 212.237.217.41: icmp_seq=3 ttl=54 time=360.514 ms

--- spd-fisrv.hostkey.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 294.590/320.340/360.514/25.228 ms
rong@KevinRongdeMacBook-Air ~ %

```

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 spd-fisrv.hostkey.com
PING spd-fisrv.hostkey.com (212.237.217.41): 56 data bytes
64 bytes from 212.237.217.41: icmp_seq=0 ttl=54 time=395.802 ms
64 bytes from 212.237.217.41: icmp_seq=1 ttl=54 time=312.455 ms
64 bytes from 212.237.217.41: icmp_seq=2 ttl=54 time=336.006 ms
64 bytes from 212.237.217.41: icmp_seq=3 ttl=54 time=354.212 ms

--- spd-fisrv.hostkey.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 312.455/349.619/395.802/30.498 ms
rong@KevinRongdeMacBook-Air ~ %

```

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 spd-fisrv.hostkey.com
PING spd-fisrv.hostkey.com (212.237.217.41): 56 data bytes
64 bytes from 212.237.217.41: icmp_seq=0 ttl=54 time=330.131 ms
64 bytes from 212.237.217.41: icmp_seq=1 ttl=54 time=349.788 ms
64 bytes from 212.237.217.41: icmp_seq=2 ttl=54 time=368.123 ms
64 bytes from 212.237.217.41: icmp_seq=3 ttl=54 time=389.968 ms

--- spd-fisrv.hostkey.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 330.131/359.582/389.968/22.133 ms
rong@KevinRongdeMacBook-Air ~ %

```

<Fig 3.2.5 3 times ping for spd-fisrv.hostkey.com>

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speed.itgate.net
PING speed.itgate.net (213.212.129.36): 56 data bytes
64 bytes from 213.212.129.36: icmp_seq=0 ttl=30 time=388.254 ms
64 bytes from 213.212.129.36: icmp_seq=1 ttl=30 time=374.819 ms
64 bytes from 213.212.129.36: icmp_seq=2 ttl=30 time=394.541 ms
64 bytes from 213.212.129.36: icmp_seq=3 ttl=30 time=389.670 ms

--- speed.itgate.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 388.254/344.621/394.541/38.355 ms
rong@KevinRongdeMacBook-Air ~ %

```

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speed.itgate.net
PING speed.itgate.net (213.212.129.36): 56 data bytes
64 bytes from 213.212.129.36: icmp_seq=0 ttl=30 time=387.373 ms
64 bytes from 213.212.129.36: icmp_seq=1 ttl=30 time=339.385 ms
64 bytes from 213.212.129.36: icmp_seq=2 ttl=30 time=358.119 ms
64 bytes from 213.212.129.36: icmp_seq=3 ttl=30 time=380.658 ms

--- speed.itgate.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 387.373/346.364/380.658/26.853 ms
rong@KevinRongdeMacBook-Air ~ %

```

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 speed.itgate.net
PING speed.itgate.net (213.212.129.36): 56 data bytes
64 bytes from 213.212.129.36: icmp_seq=0 ttl=30 time=327.636 ms
64 bytes from 213.212.129.36: icmp_seq=1 ttl=30 time=307.918 ms
64 bytes from 213.212.129.36: icmp_seq=2 ttl=30 time=366.184 ms
64 bytes from 213.212.129.36: icmp_seq=3 ttl=30 time=384.572 ms

--- speed.itgate.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 307.918/346.577/384.572/30.337 ms
rong@KevinRongdeMacBook-Air ~ %

```

<Fig 3.2.6 3 times ping for speed.itgate.net>

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 porto.speedtest.net.zon.pt
PING porto.speedtest.net.zon.pt (95.95.252.95): 56 data bytes
64 bytes from 95.95.252.95: icmp_seq=0 ttl=54 time=285.714 ms
64 bytes from 95.95.252.95: icmp_seq=1 ttl=54 time=303.376 ms
64 bytes from 95.95.252.95: icmp_seq=2 ttl=54 time=267.539 ms
64 bytes from 95.95.252.95: icmp_seq=3 ttl=54 time=342.463 ms

--- porto.speedtest.net.zon.pt ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 267.539/299.773/342.463/27.713 ms
rong@KevinRongdeMacBook-Air ~ %

```

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 porto.speedtest.net.zon.pt
PING porto.speedtest.net.zon.pt (95.95.252.95): 56 data bytes
64 bytes from 95.95.252.95: icmp_seq=0 ttl=54 time=268.110 ms
64 bytes from 95.95.252.95: icmp_seq=1 ttl=54 time=271.588 ms
64 bytes from 95.95.252.95: icmp_seq=2 ttl=54 time=347.165 ms
64 bytes from 95.95.252.95: icmp_seq=3 ttl=54 time=366.067 ms

--- porto.speedtest.net.zon.pt ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 268.110/313.233/366.067/43.912 ms
rong@KevinRongdeMacBook-Air ~ %

```

```

rong@KevinRongdeMacBook-Air ~ % ping -c 4 porto.speedtest.net.zon.pt
PING porto.speedtest.net.zon.pt (95.95.252.95): 56 data bytes
64 bytes from 95.95.252.95: icmp_seq=0 ttl=54 time=283.659 ms
64 bytes from 95.95.252.95: icmp_seq=1 ttl=54 time=306.595 ms
64 bytes from 95.95.252.95: icmp_seq=2 ttl=54 time=328.588 ms
64 bytes from 95.95.252.95: icmp_seq=3 ttl=54 time=450.399 ms

--- porto.speedtest.net.zon.pt ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 283.659/342.310/450.399/64.395 ms
rong@KevinRongdeMacBook-Air ~ %

```

<Fig 3.2.7 3 times ping for porto.speedtest.net.zon.pt>


```
rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.masnet.ec
PING speedtest.masnet.ec (170.83.216.19): 56 data bytes
64 bytes from 170.83.216.19: icmp_seq=0 ttl=34 time=430.035 ms
64 bytes from 170.83.216.19: icmp_seq=1 ttl=34 time=347.695 ms
64 bytes from 170.83.216.19: icmp_seq=2 ttl=34 time=331.479 ms
64 bytes from 170.83.216.19: icmp_seq=3 ttl=34 time=325.073 ms

--- speedtest.masnet.ec ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 325.073/358.571/430.035/42.076 ms
rong@KevinRongdeMacBook-Air ~ %
```

```
rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.masnet.ec
PING speedtest.masnet.ec (170.83.216.19): 56 data bytes
64 bytes from 170.83.216.19: icmp_seq=0 ttl=34 time=405.229 ms
64 bytes from 170.83.216.19: icmp_seq=1 ttl=34 time=424.896 ms
64 bytes from 170.83.216.19: icmp_seq=2 ttl=34 time=341.935 ms
64 bytes from 170.83.216.19: icmp_seq=3 ttl=34 time=325.178 ms

--- speedtest.masnet.ec ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 325.178/374.309/424.896/41.764 ms
rong@KevinRongdeMacBook-Air ~ %
```

```
rong@KevinRongdeMacBook-Air ~ % ping -c 4 speedtest.masnet.ec
PING speedtest.masnet.ec (170.83.216.19): 56 data bytes
64 bytes from 170.83.216.19: icmp_seq=0 ttl=34 time=325.260 ms
64 bytes from 170.83.216.19: icmp_seq=1 ttl=34 time=337.505 ms
64 bytes from 170.83.216.19: icmp_seq=2 ttl=34 time=325.110 ms
64 bytes from 170.83.216.19: icmp_seq=3 ttl=34 time=325.243 ms

--- speedtest.masnet.ec ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 325.110/328.279/337.505/5.327 ms
rong@KevinRongdeMacBook-Air ~ %
```

<Fig 3.2.8 3 times ping for speedtest.masnet.ec>

```
rong@KevinRongdeMacBook-Air ~ % ping -c 4 dal.speedtest.clouvider.net
PING dal.speedtest.clouvider.net (2.56.188.136): 56 data bytes
64 bytes from 2.56.188.136: icmp_seq=0 ttl=47 time=191.497 ms
64 bytes from 2.56.188.136: icmp_seq=1 ttl=47 time=195.053 ms
64 bytes from 2.56.188.136: icmp_seq=2 ttl=47 time=191.212 ms
64 bytes from 2.56.188.136: icmp_seq=3 ttl=47 time=190.588 ms

--- dal.speedtest.clouvider.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 190.588/192.087/195.053/1.743 ms
rong@KevinRongdeMacBook-Air ~ %
```

```
rong@KevinRongdeMacBook-Air ~ % ping -c 4 dal.speedtest.clouvider.net
PING dal.speedtest.clouvider.net (2.56.188.136): 56 data bytes
64 bytes from 2.56.188.136: icmp_seq=0 ttl=47 time=195.526 ms
64 bytes from 2.56.188.136: icmp_seq=1 ttl=47 time=196.429 ms
64 bytes from 2.56.188.136: icmp_seq=2 ttl=47 time=196.230 ms
64 bytes from 2.56.188.136: icmp_seq=3 ttl=47 time=192.074 ms

--- dal.speedtest.clouvider.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 192.074/195.065/196.429/1.759 ms
rong@KevinRongdeMacBook-Air ~ %
```

```
rong@KevinRongdeMacBook-Air ~ % ping -c 4 dal.speedtest.clouvider.net
PING dal.speedtest.clouvider.net (2.56.188.136): 56 data bytes
64 bytes from 2.56.188.136: icmp_seq=0 ttl=47 time=192.013 ms
64 bytes from 2.56.188.136: icmp_seq=1 ttl=47 time=192.488 ms
64 bytes from 2.56.188.136: icmp_seq=2 ttl=47 time=192.534 ms
64 bytes from 2.56.188.136: icmp_seq=3 ttl=47 time=190.827 ms

--- dal.speedtest.clouvider.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 190.827/191.965/192.534/0.688 ms
rong@KevinRongdeMacBook-Air ~ %
```

<Fig 3.2.9 3 times ping for dal.speedtest.clouvider.net>

```
rong@KevinRongdeMacBook-Air ~ % ping -c 4 iperf.he.net
PING 1500.mtu.he.net (216.218.207.42): 56 data bytes
64 bytes from 216.218.207.42: icmp_seq=0 ttl=50 time=453.270 ms
64 bytes from 216.218.207.42: icmp_seq=1 ttl=50 time=475.908 ms
64 bytes from 216.218.207.42: icmp_seq=2 ttl=50 time=296.809 ms
64 bytes from 216.218.207.42: icmp_seq=3 ttl=50 time=315.623 ms

--- 1500.mtu.he.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 296.809/385.403/475.908/79.867 ms
rong@KevinRongdeMacBook-Air ~ %
```

```
rong@KevinRongdeMacBook-Air ~ % ping -c 4 iperf.he.net
PING 1500.mtu.he.net (216.218.207.42): 56 data bytes
64 bytes from 216.218.207.42: icmp_seq=0 ttl=50 time=341.667 ms
64 bytes from 216.218.207.42: icmp_seq=1 ttl=50 time=287.294 ms
64 bytes from 216.218.207.42: icmp_seq=2 ttl=50 time=289.169 ms
64 bytes from 216.218.207.42: icmp_seq=3 ttl=50 time=295.491 ms

--- 1500.mtu.he.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 287.294/303.400/341.667/22.290 ms
rong@KevinRongdeMacBook-Air ~ %
```

```
rong@KevinRongdeMacBook-Air ~ % ping -c 4 iperf.he.net
PING 1500.mtu.he.net (216.218.207.42): 56 data bytes
64 bytes from 216.218.207.42: icmp_seq=0 ttl=50 time=282.754 ms
64 bytes from 216.218.207.42: icmp_seq=1 ttl=50 time=315.623 ms
64 bytes from 216.218.207.42: icmp_seq=2 ttl=50 time=287.835 ms
64 bytes from 216.218.207.42: icmp_seq=3 ttl=50 time=353.514 ms

--- 1500.mtu.he.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 282.754/309.932/353.514/28.101 ms
rong@KevinRongdeMacBook-Air ~ %
```

<Fig 3.2.10 3 times ping for iperf.he.net>

Section 4

```
rong --zsh - 120x48
Last login: Thu Apr 23 22:44:58 on ttys000
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.awbb.net -p 80
Client connecting to speedtest.awbb.net, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58646 connected with 184.245.79.238 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  85.8 MBytes  71.3 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.awbb.net -p 80
Client connecting to speedtest.awbb.net, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58658 connected with 184.245.79.238 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  48.8 MBytes  40.9 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.awbb.net -p 80
Client connecting to speedtest.awbb.net, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58659 connected with 184.245.79.238 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  86.8 MBytes  72.1 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
```

<4.1(a) speedtest.awbb.net>

```
rong --zsh - 120x48
rong@KevinRongdeMacBook-Air ~ % iperf -c spd-desrv.hostkey.com -p 80
Client connecting to spd-desrv.hostkey.com, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58695 connected with 185.84.224.68 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  185 MBytes  89.3 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c spd-desrv.hostkey.com -p 80
Client connecting to spd-desrv.hostkey.com, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58698 connected with 185.84.224.68 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  181 MBytes  84.5 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c spd-desrv.hostkey.com -p 80
Client connecting to spd-desrv.hostkey.com, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58704 connected with 185.84.224.68 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.02 sec  86.5 MBytes  62.5 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
```

<4.1(b) spd-desrv.hostkey.com>

```
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.wobcom.de -p 80
Client connecting to speedtest.wobcom.de, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58728 connected with 62.176.224.217 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.00 sec  86.6 MBytes  72.6 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.wobcom.de -p 80
Client connecting to speedtest.wobcom.de, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58731 connected with 62.176.224.217 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  93.1 MBytes  78.1 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.wobcom.de -p 80
Client connecting to speedtest.wobcom.de, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58736 connected with 62.176.224.217 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  98.9 MBytes  76.2 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
```

<4.1(c) speedtest.wobcom.de>


```
rong -- zsh -- 120x48
rong@KevinRongdeMacBook-Air ~ % iperf -c rychlost.poda.cz -p 80
Client connecting to rychlost.poda.cz, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58767 connected with 85.135.13.2 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.02 sec  49.4 MBytes  58.3 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c rychlost.poda.cz -p 80
Client connecting to rychlost.poda.cz, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58769 connected with 85.135.13.2 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.00 sec  64.4 MBytes  54.9 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c rychlost.poda.cz -p 80
Client connecting to rychlost.poda.cz, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58774 connected with 85.135.13.2 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.02 sec  4.76 MBytes  3.98 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
```

<4.1(d) rychlost.poda.cz>

```
rong -- zsh -- 120x48
rong@KevinRongdeMacBook-Air ~ % iperf -c spd-fisrv.hostkey.com -p 80
Client connecting to spd-fisrv.hostkey.com, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58792 connected with 212.237.217.41 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.03 sec  91.6 MBytes  74.9 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c spd-fisrv.hostkey.com -p 80
Client connecting to spd-fisrv.hostkey.com, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58799 connected with 212.237.217.41 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.00 sec  64.4 MBytes  54.9 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c spd-fisrv.hostkey.com -p 80
Client connecting to spd-fisrv.hostkey.com, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58808 connected with 212.237.217.41 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.03 sec  51.5 MBytes  43.1 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
```

<4.1(e) spd-fisrv.hostkey.com>

```
rong -- zsh -- 120x48
rong@KevinRongdeMacBook-Air ~ % iperf -c speed.itgate.net -p 80
Client connecting to speed.itgate.net, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58849 connected with 213.212.129.36 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-11.26 sec  1.50 MBytes  1.12 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c speed.itgate.net -p 80
Client connecting to speed.itgate.net, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58841 connected with 213.212.129.36 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.06 sec  17.6 MBytes  14.7 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c speed.itgate.net -p 80
Client connecting to speed.itgate.net, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58846 connected with 213.212.129.36 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  100 MBytes  84.9 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
```

<4.1(f) speed.itgate.net>

```
rong --zsh - 120x48
rong@KevinRongdeMacBook-Air ~ % iperf -c porto.speedtest.net.zon.pt -p 80
Client connecting to porto.speedtest.net.zon.pt, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58861 connected with 95.95.252.95 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.04 sec  16.6 MBytes  13.9 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c porto.speedtest.net.zon.pt -p 80
Client connecting to porto.speedtest.net.zon.pt, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58863 connected with 95.95.252.95 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.14 sec  16.5 MBytes  13.7 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c porto.speedtest.net.zon.pt -p 80
Client connecting to porto.speedtest.net.zon.pt, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58866 connected with 95.95.252.95 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.05 sec  16.4 MBytes  13.7 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
```

<4.1(g) porto.speedtest.net.zon.pt>

```
rong --zsh - 120x48
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.masnet.ec -p 80
Client connecting to speedtest.masnet.ec, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 60241 connected with 170.83.216.19 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.34 sec  16.8 MBytes  11.6 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.masnet.ec -p 80
Client connecting to speedtest.masnet.ec, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 60245 connected with 170.83.216.19 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.34 sec  23.0 MBytes  18.7 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.masnet.ec -p 80
Client connecting to speedtest.masnet.ec, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 60248 connected with 170.83.216.19 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.34 sec  15.5 MBytes  12.6 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
```

<4.1(h) speedtest.masnet.ec>

```
rong --zsh - 120x48
rong@KevinRongdeMacBook-Air ~ % iperf -c dal.speedtest.clouvider.net -p 80
Client connecting to dal.speedtest.clouvider.net, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 58928 connected with 2.56.188.136 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  95.4 MBytes  79.9 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c dal.speedtest.clouvider.net -p 80
Client connecting to dal.speedtest.clouvider.net, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 59061 connected with 2.56.188.136 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.00 sec  123 MBytes  103 Mbits/sec
rong@KevinRongdeMacBook-Air ~ % iperf -c dal.speedtest.clouvider.net -p 80
Client connecting to dal.speedtest.clouvider.net, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 59005 connected with 2.56.188.136 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.02 sec  133 MBytes  111 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %
```

<4.1(i) dal.speedtest.clouvider.net>

```

rong@KevinRongdeMacBook-Air ~ % iperf3 -c iperf.he.net -R
Connecting to host iperf.he.net, port 5201
Reverse mode, remote host iperf.he.net is sending
[ 7] local 172.16.145.22 port 60442 connected to 216.218.207.42 port 5201
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-1.01 sec  128 KBytes  1.04 Mbits/sec
[ 7] 1.01-2.01 sec  1.38 MBytes  11.6 Mbits/sec
[ 7] 2.01-3.00 sec  5.62 MBytes  47.2 Mbits/sec
[ 7] 3.00-4.01 sec  5.50 MBytes  46.1 Mbits/sec
[ 7] 4.01-5.00 sec  6.62 MBytes  55.8 Mbits/sec
[ 7] 5.00-6.00 sec  8.25 MBytes  69.1 Mbits/sec
[ 7] 6.00-7.00 sec  5.50 MBytes  46.1 Mbits/sec
[ 7] 7.00-8.01 sec  1.50 MBytes  12.6 Mbits/sec
[ 7] 8.01-9.00 sec  4.00 MBytes  33.6 Mbits/sec
[ 7] 9.00-10.01 sec  4.50 MBytes  37.7 Mbits/sec
-----
[ ID] Interval      Transfer    Bitrate      Retr      sender      receiver
[ 7] 0.00-10.30 sec  46.4 MBytes  37.8 Mbits/sec  1694
[ 7] 0.00-10.01 sec  43.0 MBytes  36.1 Mbits/sec

iperf Done.

rong@KevinRongdeMacBook-Air ~ % iperf3 -c iperf.he.net -R
Connecting to host iperf.he.net, port 5201
Reverse mode, remote host iperf.he.net is sending
[ 7] local 172.16.145.22 port 60494 connected to 216.218.207.42 port 5201
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-1.01 sec  128 KBytes  1.04 Mbits/sec
[ 7] 1.01-2.00 sec  1.12 MBytes  9.46 Mbits/sec
[ 7] 2.00-3.00 sec  2.12 MBytes  17.8 Mbits/sec
[ 7] 3.00-4.00 sec  1.62 MBytes  13.6 Mbits/sec
[ 7] 4.00-5.00 sec  2.12 MBytes  17.8 Mbits/sec
[ 7] 5.00-6.00 sec  1.75 MBytes  14.7 Mbits/sec
[ 7] 6.00-7.00 sec  2.12 MBytes  17.8 Mbits/sec
[ 7] 7.00-8.00 sec  2.12 MBytes  17.8 Mbits/sec
[ 7] 8.00-9.01 sec  2.50 MBytes  21.0 Mbits/sec
[ 7] 9.01-10.01 sec  2.75 MBytes  23.1 Mbits/sec
-----
[ ID] Interval      Transfer    Bitrate      Retr      sender      receiver
[ 7] 0.00-10.29 sec  21.0 MBytes  17.1 Mbits/sec      1
[ 7] 0.00-10.01 sec  18.4 MBytes  15.4 Mbits/sec

iperf Done.
rong@KevinRongdeMacBook-Air ~ %

```

<4.1(j) iperf.he.net>

```

rong -- -zsh -- 120x49
rong@KevinRongdeMacBook-Air ~ % sudo traceroute -I speedtest.masnet.ec
traceroute to speedtest.masnet.ec (170.83.216.19), 64 hops max, 48 byte packets
 1 172.16.145.1 (172.16.145.1) 3.527 ms 3.239 ms 3.163 ms
 2 103.131.13.193 (103.131.13.193) 16.181 ms 18.908 ms 164.277 ms
 3 103.251.4.80 (103.251.4.80) 3.885 ms 3.962 ms 3.104 ms
 4 po2.nxtp01.mel02.nexthop.net.au (103.251.4.31) 3.368 ms 3.235 ms 3.183 ms
 5 202.139.7.25 (202.139.7.25) 4.593 ms 3.815 ms 3.649 ms
 6 ***
 7 ***
 8 be3257.ccr01.sin01.atlas.cogentco.com (154.54.140.45) 200.810 ms * *
 9 be9040.ccr01.lax04.atlas.cogentco.com (154.54.87.145) 200.595 ms 201.029 ms 200.327 ms
10 be3360.ccr02.lax01.atlas.cogentco.com (154.54.25.149) 200.657 ms 199.926 ms 200.762 ms
11 be2932.ccr32.phx01.atlas.cogentco.com (154.54.45.161) 219.358 ms 218.938 ms 219.397 ms
12 be5473.ccr22.elp02.atlas.cogentco.com (154.54.166.69) 217.982 ms 217.817 ms 217.930 ms
13 be3846.ccr32.dfw01.atlas.cogentco.com (154.54.165.29) 241.939 ms 242.163 ms 241.728 ms
14 port-channel18122.ccr72-jan02.atlas.cogentco.com (154.54.41.54) 258.761 ms 257.852 ms 257.998 ms
15 be0473.ccr02.mia03.atlas.cogentco.com (154.54.5.162) 272.846 ms 273.648 ms 272.040 ms
16 be2258.ccr01.mia03.atlas.cogentco.com (154.54.168.85) 272.938 ms 273.707 ms 273.444 ms
17 38.104.94.178 (38.104.94.178) 273.869 ms 273.970 ms 273.250 ms
18 ***
19 ***
20 190.242.173.79 (190.242.173.79) 316.895 ms 309.120 ms *
21 ***
22 ***
23 ***
24 voip.masnet.ec (170.83.216.5) 327.428 ms 327.151 ms 327.581 ms
25 speedtest.masnet.ec (170.83.216.19) 328.622 ms 328.123 ms 327.314 ms
rong@KevinRongdeMacBook-Air ~ %

rong -- -zsh -- 120x49
rong@KevinRongdeMacBook-Air ~ % iperf -c speedtest.masnet.ec -p 80
Client connecting to speedtest.masnet.ec, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 62654 connected with 170.83.216.19 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.34 sec  25.4 MBytes  20.6 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %

```

<4.2.1(a) Ecuador (Closest) – morning >

```
rong -- -zsh -- 120x49
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月 24日 星期五 13时20分08秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ % sudo traceroute -I speedtest.masnet.ec
traceroute to speedtest.masnet.ec (170.83.216.19), 64 hops max, 48 byte packets
 1 172.16.145.1 (172.16.145.1) 3.472 ms 3.288 ms 3.121 ms
 2 183.131.13.193 (183.131.13.193) 20.346 ms 21.495 ms 12.542 ms
 3 183.251.4.80 (183.251.4.80) 3.189 ms 3.376 ms 3.157 ms
 4 po2.nxtpe81.me102.nexthop.net.au (183.251.4.31) 3.195 ms 2.543 ms 3.241 ms
 5 202.139.7.25 (202.139.7.25) 3.550 ms 4.105 ms 4.242 ms
 6 * * *
 7 * * *
 8 be3257.ccr81.sin01.atlas.cogentco.com (154.54.140.45) 199.481 ms 199.115 ms *
 9 be9040.ccr41.lax04.atlas.cogentco.com (154.54.87.145) 204.415 ms 200.364 ms 200.628 ms
10 be3360.ccr42.lax01.atlas.cogentco.com (154.54.25.149) 200.070 ms 200.541 ms 200.512 ms
11 be2932.ccr32.phx01.atlas.cogentco.com (154.54.45.161) 246.953 ms 219.378 ms 219.856 ms
12 be5473.ccr22.elp02.atlas.cogentco.com (154.54.166.69) 218.215 ms 217.700 ms 218.172 ms
13 be3846.ccr32.dfw01.atlas.cogentco.com (154.54.165.29) 320.404 ms 241.799 ms 242.094 ms
14 port-channel18122.ccr92.jan02.atlas.cogentco.com (154.54.41.54) 320.265 ms 296.493 ms 258.599 ms
15 * be8673.ccr02.mia03.atlas.cogentco.com (154.54.5.162) 272.317 ms 270.347 ms
16 be2258.ccr41.mia03.atlas.cogentco.com (154.54.168.85) 271.541 ms 282.240 ms 277.338 ms
17 38.104.94.178 (38.104.94.178) 287.592 ms 343.661 ms 306.995 ms
18 * * *
19 * * *
20 198.242.173.79 (198.242.173.79) 317.086 ms * 309.000 ms
21 * * *
22 * * *
23 * * *
24 volp.masnet.ec (170.83.216.5) 329.039 ms 339.233 ms 326.434 ms
25 speedtest.masnet.ec (170.83.216.19) 324.647 ms 325.383 ms 324.975 ms
rong@KevinRongdeMacBook-Air-2 ~ %
```

```
rong -- -zsh -- 120x49
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月 24日 星期五 13时18分20秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ % iperf -c speedtest.masnet.ec -p 80
Client connecting to speedtest.masnet.ec, TCP port 80
TCP window size: 128 KByte (default)
-----
[ 1] local 172.16.145.22 port 64336 connected with 170.83.216.19 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.34 sec 27.4 MBytes 22.2 Mbits/sec
rong@KevinRongdeMacBook-Air-2 ~ %
```

<4.2.1(b) Ecuador (Closest) – afternoon >

```
rong@KevinRongdeMacBook-Air-2 ~ % sudo traceroute -I speedtest.masnet.ec
Password:
traceroute to speedtest.masnet.ec (170.83.216.19), 64 hops max, 48 byte packets
 1 172.16.145.1 (172.16.145.1) 3.040 ms 2.608 ms 3.226 ms
 2 183.131.13.193 (183.131.13.193) 8.446 ms 20.797 ms 18.072 ms
 3 183.251.4.80 (183.251.4.80) 3.028 ms 6.045 ms 3.048 ms
 4 po2.nxtpe81.me102.nexthop.net.au (183.251.4.31) 3.167 ms 3.592 ms 3.191 ms
 5 202.139.7.25 (202.139.7.25) 4.302 ms 3.917 ms 3.966 ms
 6 * * *
 7 * * *
 8 * be3257.ccr81.sin01.atlas.cogentco.com (154.54.140.45) 295.558 ms 199.387 ms
 9 be9040.ccr41.lax04.atlas.cogentco.com (154.54.87.145) 200.559 ms 200.773 ms 200.688 ms
10 be3360.ccr42.lax01.atlas.cogentco.com (154.54.25.149) 200.623 ms 200.023 ms 200.518 ms
11 be2932.ccr32.phx01.atlas.cogentco.com (154.54.45.161) 220.051 ms 220.448 ms 219.510 ms
12 be5473.ccr22.elp02.atlas.cogentco.com (154.54.166.69) 217.317 ms 216.899 ms 216.203 ms
13 be3846.ccr32.dfw01.atlas.cogentco.com (154.54.165.29) 242.303 ms 242.073 ms 242.273 ms
14 port-channel18122.ccr92.jan02.atlas.cogentco.com (154.54.41.54) 261.754 ms 257.740 ms 317.204 ms
15 be8673.ccr02.mia03.atlas.cogentco.com (154.54.5.162) 274.579 ms 272.441 ms 273.913 ms
16 be2258.ccr41.mia03.atlas.cogentco.com (154.54.168.85) 274.154 ms 274.127 ms 274.312 ms
17 38.104.94.178 (38.104.94.178) 273.805 ms 273.889 ms 276.073 ms
18 * * *
19 * * *
20 198.242.173.79 (198.242.173.79) 300.778 ms 309.086 ms 300.613 ms
21 * * *
22 * * *
23 * * *
24 volp.masnet.ec (170.83.216.5) 327.209 ms 320.309 ms 320.004 ms
25 speedtest.masnet.ec (170.83.216.19) 326.995 ms 328.176 ms 327.354 ms
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月 24日 星期五 18时13分03秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ %
```

```
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月 24日 星期五 18时09分19秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ % iperf -c speedtest.masnet.ec -p 80
Client connecting to speedtest.masnet.ec, TCP port 80
TCP window size: 128 KByte (default)
-----
[ 1] local 172.16.145.22 port 50078 connected with 170.83.216.19 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.34 sec 30.1 MBytes 24.4 Mbits/sec
rong@KevinRongdeMacBook-Air-2 ~ %
```

<4.2.1(c) Ecuador (Closest) – evening >

```

rong@KevinRongdeMacBook-Air-2 ~ % sudo traceroute -I speedtest.masnet.ec
traceroute to speedtest.masnet.ec (170.83.216.19): 64 hops max, 48 byte packets
 1 172.16.145.1 (172.16.145.1) 7.843 ms 2.848 ms 3.149 ms
 2 103.131.13.193 (103.131.13.193) 16.043 ms 20.619 ms 23.448 ms
 3 103.251.4.80 (103.251.4.80) 3.809 ms 2.975 ms 3.174 ms
 4 po2.nxtpe01.me102.nexthop.net.au (103.251.4.31) 3.174 ms 3.261 ms 3.312 ms
 5 202.139.7.25 (202.139.7.25) 19.321 ms 4.131 ms 3.662 ms
 6 * * *
 7 * * *
 8 * ba3257.ccr81.sin01.atlas.cogentco.com (154.54.140.45) 204.050 ms 200.580 ms
 9 be9840.ccr41.lax04.atlas.cogentco.com (154.54.87.145) 200.959 ms 200.541 ms 200.636 ms
10 be3308.ccr42.lax01.atlas.cogentco.com (154.54.25.149) 200.952 ms 199.680 ms 200.826 ms
11 be2932.ccr32.ph01.atlas.cogentco.com (154.54.45.161) 228.293 ms 219.616 ms 311.757 ms
12 be5473.ccr22.elp02.atlas.cogentco.com (154.54.166.69) 308.780 ms 294.666 ms 218.240 ms
13 be3846.ccr32.dfw01.atlas.cogentco.com (154.54.165.29) 242.712 ms 446.278 ms 307.238 ms
14 port-channel18122.ccr92.jan02.atlas.cogentco.com (154.54.41.54) 304.801 ms 299.792 ms 307.299 ms
15 * * *
16 be2258.ccr41.mia03.atlas.cogentco.com (154.54.168.85) 350.235 ms 341.800 ms 306.923 ms
17 36.104.94.178 (36.104.94.178) 307.453 ms 293.445 ms 307.732 ms
18 * * *
19 * * *
20 190.242.173.79 (190.242.173.79) 309.041 ms 336.014 ms 375.729 ms
21 * * *
22 * * *
23 * * *
24 voip.masnet.ec (170.83.216.5) 524.868 ms 327.077 ms 388.391 ms
25 speedtest.masnet.ec (170.83.216.19) 326.676 ms 328.000 ms 326.504 ms
rong@KevinRongdeMacBook-Air-2 ~ %

```

```

rong -- -zsh -- 120x49

rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月24日 星期五 23时06分08秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ % iperf -c speedtest.masnet.ec -p 80

Client connecting to speedtest.masnet.ec, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 58942 connected with 170.83.216.19 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.34 sec  25.6 MBytes  20.8 Mbits/sec
rong@KevinRongdeMacBook-Air-2 ~ %

```

<4.2.1(d) Ecuador (Closest) – night >

```

rong -- -zsh -- 120x49

rong@KevinRongdeMacBook-Air ~ % iperf -c porto.speedtest.net.zon.pt -p 80

Client connecting to porto.speedtest.net.zon.pt, TCP port 80
TCP window size: 128 KByte (default)

[ 1] local 172.16.145.22 port 62608 connected with 95.95.252.95 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.03 sec  16.5 MBytes  13.8 Mbits/sec
rong@KevinRongdeMacBook-Air ~ %

```

```

rong -- -zsh -- 120x49

rong@KevinRongdeMacBook-Air ~ % sudo traceroute -I porto.speedtest.net.zon.pt
traceroute to porto.speedtest.net.zon.pt (95.95.252.95): 64 hops max, 48 byte packets
 1 172.16.145.1 (172.16.145.1) 3.949 ms 3.377 ms 3.146 ms
 2 103.131.13.193 (103.131.13.193) 22.511 ms 20.360 ms 22.641 ms
 3 103.251.4.80 (103.251.4.80) 6.488 ms 4.114 ms 2.484 ms
 4 po2.nxtpe01.me102.nexthop.net.au (103.251.4.31) 3.134 ms 2.780 ms 3.157 ms
 5 202.139.7.25 (202.139.7.25) 3.909 ms 4.079 ms 3.157 ms
 6 * * *
 7 * * *
 8 172.26.11.185 (172.26.11.185) 51.896 ms 52.222 ms 52.129 ms
 9 * * *
10 be3077.ccr31.bio02.atlas.cogentco.com (154.54.39.225) 251.437 ms 252.193 ms 251.179 ms
11 be2313.rcr51.opo01.atlas.cogentco.com (154.54.61.102) 263.984 ms 263.331 ms 264.449 ms
12 be2444.rcr21.lis01.atlas.cogentco.com (154.54.61.213) 263.219 ms 261.759 ms 262.742 ms
13 * * *
14 * * *
15 porto.speedtest.net.zon.pt (95.95.252.95) 269.434 ms 269.557 ms 270.068 ms
rong@KevinRongdeMacBook-Air ~ %

```

<4.2.2(a) Portugal (Furthest) – morning >


```
rong -- -zsh -- 120x49
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月24日 星期五 13时15分39秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ % sudo traceroute -I porto.speedtest.net.zon.pt
Password:
traceroute to porto.speedtest.net.zon.pt (95.95.252.95), 64 hops max, 48 byte packets
 1 172.16.145.1 (172.16.145.1) 3.511 ms 3.123 ms 3.028 ms
 2 103.131.13.193 (103.131.13.193) 24.623 ms 14.908 ms 13.894 ms
 3 103.251.4.80 (103.251.4.80) 3.139 ms 2.688 ms 3.134 ms
 4 po2.nxtpe01.me102.nextop.net.au (103.251.4.31) 3.174 ms 3.351 ms 3.191 ms
 5 202.139.7.25 (202.139.7.25) 6.888 ms 19.646 ms 15.035 ms
 6 ***
 7 ***
 8 172.26.11.185 (172.26.11.185) 52.221 ms 52.719 ms 52.441 ms
 9 ***
10 be3077.ccr31.bio02.atlas.cogentco.com (154.54.39.225) 251.512 ms 289.561 ms 251.509 ms
11 be2313.rcr51.opo01.atlas.cogentco.com (154.54.61.102) 265.420 ms 265.194 ms 264.208 ms
12 be2444.rcr21.lis01.atlas.cogentco.com (154.54.61.213) 262.341 ms 262.085 ms 262.322 ms
13 ***
14 ***
15 porto.speedtest.net.zon.pt (95.95.252.95) 296.088 ms 268.913 ms 269.663 ms
rong@KevinRongdeMacBook-Air-2 ~ %
```

```
rong -- -zsh -- 120x49
Last login: Fri Apr 24 09:25:44 on ttys001
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月24日 星期五 13时15分46秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ % iperf -c porto.speedtest.net.zon.pt -p 80

Client connecting to porto.speedtest.net.zon.pt, TCP port 80
TCP window size: 128 KByte (default)
-----
[ 1] local 172.16.145.22 port 64040 connected with 95.95.252.95 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.03 sec  16.0 MBytes  13.4 Mbits/sec
rong@KevinRongdeMacBook-Air-2 ~ %
```

<4.2.2(b) Portugal (Furthest) – afternoon >

```
rong -- -zsh -- 120x49
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月24日 星期五 18时04分19秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ % sudo traceroute -I porto.speedtest.net.zon.pt
Password:
traceroute to porto.speedtest.net.zon.pt (95.95.252.95), 64 hops max, 48 byte packets
 1 172.16.145.1 (172.16.145.1) 2.743 ms 3.369 ms 3.113 ms
 2 103.131.13.193 (103.131.13.193) 37.662 ms 21.141 ms 89.342 ms
 3 103.251.4.80 (103.251.4.80) 2.834 ms 3.826 ms 3.151 ms
 4 po2.nxtpe01.me102.nextop.net.au (103.251.4.31) 2.786 ms 4.205 ms 3.595 ms
 5 202.139.7.25 (202.139.7.25) 4.852 ms 6.654 ms 5.099 ms
 6 ***
 7 ***
 8 172.26.11.185 (172.26.11.185) 52.367 ms 52.189 ms 52.547 ms
 9 ***
10 be3077.ccr31.bio02.atlas.cogentco.com (154.54.39.225) 263.278 ms 250.663 ms 250.526 ms
11 be2313.rcr51.opo01.atlas.cogentco.com (154.54.61.102) 262.807 ms 265.884 ms 270.168 ms
12 be2444.rcr21.lis01.atlas.cogentco.com (154.54.61.213) 261.133 ms 261.731 ms 260.804 ms
13 ***
14 ***
15 porto.speedtest.net.zon.pt (95.95.252.95) 268.343 ms 268.293 ms 268.393 ms
rong@KevinRongdeMacBook-Air-2 ~ %
```

```
rong -- -zsh -- 120x49
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月24日 星期五 18时02分41秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ % iperf -c porto.speedtest.net.zon.pt -p 80

Client connecting to porto.speedtest.net.zon.pt, TCP port 80
TCP window size: 128 KByte (default)
-----
[ 1] local 172.16.145.22 port 49945 connected with 95.95.252.95 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.01 sec  16.5 MBytes  13.8 Mbits/sec
rong@KevinRongdeMacBook-Air-2 ~ %
```

<4.2.2(c) Portugal (Furthest) – evening >


```
rong@KevinRongdeMacBook-Air-2 ~ % sudo traceroute -I porto.speedtest.net.zon.pt
Password:
traceroute to porto.speedtest.net.zon.pt (95.95.252.95), 64 hops max, 48 byte packets
 1 172.16.145.1 (172.16.145.1) 3.400 ms 2.819 ms 3.178 ms
 2 103.131.13.193 (103.131.13.193) 1083.512 ms 57.163 ms 24.486 ms
 3 103.251.4.80 (103.251.4.80) 2.975 ms 3.797 ms 3.153 ms
 4 po2.nxtpe01.mel02.nextop.net.au (103.251.4.31) 3.126 ms 2.642 ms 3.692 ms
 5 202.139.7.25 (202.139.7.25) 4.311 ms 3.801 ms 3.275 ms
 6 * * *
 7 * * *
 8 172.26.11.185 (172.26.11.185) 55.642 ms 51.912 ms 52.350 ms
 9 * * *
10 be3077.ccr31.bio02.atlas.cogentco.com (154.54.39.225) 306.137 ms 294.238 ms 306.780 ms
11 be2313.rcr51.opo01.atlas.cogentco.com (154.54.61.102) 307.885 ms 262.104 ms 262.211 ms
12 be2444.rcr21.lis01.atlas.cogentco.com (154.54.61.213) 282.910 ms 296.246 ms 306.321 ms
13 * * *
14 * * *
15 porto.speedtest.net.zon.pt (95.95.252.95) 396.138 ms 301.829 ms 306.821 ms
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月24日 星期五 23时05分27秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ %
```

```
rong -- -zsh -- 120x49
rong@KevinRongdeMacBook-Air-2 ~ % date
2026年 4月24日 星期五 23时02分56秒 AEST
rong@KevinRongdeMacBook-Air-2 ~ % iperf -c porto.speedtest.net.zon.pt -p 80
Client connecting to porto.speedtest.net.zon.pt, TCP port 80
TCP window size: 128 KByte (default)
[ 1] local 172.16.145.22 port 50873 connected with 95.95.252.95 port 80
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.00-10.00 sec  16.3 MBytes  13.6 Mbits/sec
rong@KevinRongdeMacBook-Air-2 ~ %
```

<4.2.2(d) Portugal (Furthest) – night >